> **Draft v0.1 · March 2026**
>
> This is the first draft of our GPU vs TPU benchmarking workshop. What began as a scoping exercise evolved into a deeper investigation than initially expected. The result is an introductory-to-intermediate workshop grounded in real hardware measurements, transparent about trade-offs, and structured around practical decision rules practitioners can apply.
>
> The workshop currently includes six sessions covering batch scaling, sequence length, model depth, graph compilation constraints, precision (FP32/FP16/BF16), and framework interoperability (PyTorch vs JAX). Each session produced at least one non-obvious result that helped shape the analysis.
>
> **A few caveats:** - This is a first draft. Roughly 70% of the planned content is complete and a few known issues are being addressed. - All benchmarks were run on low-cost GCP instances (NVIDIA L4 and a single-chip TPU v5litepod). We do not currently have access to larger hardware such as A100/H100 GPUs, multi-GPU or multi-node setups, or larger TPU pods, which would allow a more comprehensive analysis. - INT8 and INT4 benchmarks are still pending.
>
> **Feedback welcome:** - Does this direction make sense for the workshop? - Would benchmarking full end-to-end workloads (e.g., BERT or GPT-2 training) be more useful? - Is there a way to access larger hardware, or someone we should speak with?
>
> A PDF summary of the key findings and a zip of the full repository are available on request. Direct GitHub access can also be arranged — just send a username.
>
> Work on the remaining items will resume next week. In the meantime, any feedback is very welcome.

Workshop Introduction: GPU vs TPU for Transformer Workloads

What this workshop is

This workshop benchmarks a single Transformer encoder block systematically across two cloud accelerators — an NVIDIA L4 GPU and a Google TPU v5e — to answer a practical question: **when should you pick one over the other, and by how much does it matter?**

The benchmark is intentionally small. One encoder block is a controlled proxy: it isolates the hardware and software variables without the noise of a full model pipeline. The findings scale to real workloads — the same memory-bandwidth ceiling, the same compilation model, the same precision trade-offs apply at BERT-base or GPT-2 scale.

Why these two accelerators?

The hardware

Property	NVIDIA L4 (GPU)	Google TPU v5e (v5litepod-1)
Architecture	Ada Lovelace (2022)	Systolic array MXU
Memory	23.7 GB GDDR6	16 GB HBM2
Memory bandwidth	~300 GB/s	~819 GB/s
Peak compute (BF16)	~121 TFLOPS	~393 TFLOPS
Peak compute (INT8)	~242 TOPS	~786 TOPS
TDP	72 W	—
GCP hourly rate (us-central1, early 2026)	~$0.70 / hr	~$1.20 / hr

On paper the TPU has a 3.25× compute advantage and a 2.7× memory bandwidth advantage — for a 1.71× price premium. This asymmetry is not a problem with the comparison; it is the central question of the workshop. The goal is to find out under which conditions that hardware advantage actually translates into cheaper, faster training.

Why the L4, not an A100?

The L4 is GCP's most accessible modern GPU for training workloads. The next step up — an A100 — is roughly 3–5× more expensive per hour and places the comparison in a different budget tier. The L4 is what a cost-conscious practitioner reaches for first.

Why the TPU v5e, not a cheaper option?

The TPU v5e single-chip (v5litepod-1) is the cheapest TPU available on GCP. There is no sub-$1.20/hr TPU option. Older generations (v2, v3) are being deprecated. The v5e is the realistic entry point.

The comparison is honest

The L4 is inference-optimised — lower power, lower TFLOPS than a training GPU of the same generation. That is acknowledged throughout the workshop. The comparison is not "best GPU vs best TPU" — it is "the GPU you would most likely spin up on GCP vs the TPU you would most likely spin up on GCP, at their listed prices."

The right comparison metric is not raw throughput but **samples per dollar**. The workshop keeps both in view.

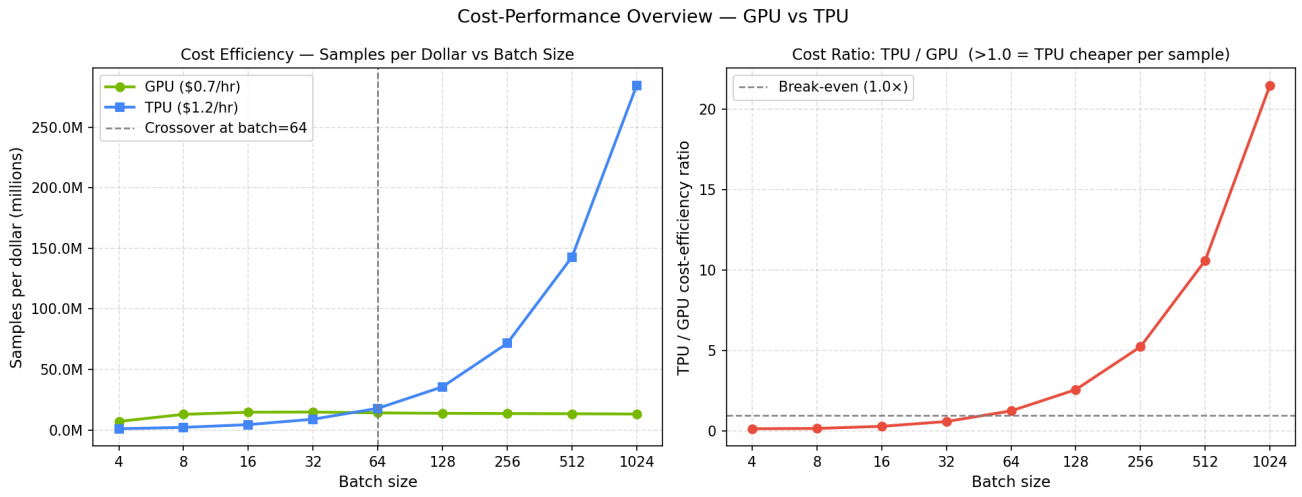
The economic context

The cost analysis established the following using FP32 training throughput from Session 1:

Batch	GPU (samples/$)	TPU (samples/$)	TPU cheaper?
4	7.1M	1.1M	No
8	13.0M	2.2M	No
16	14.7M	4.5M	No
32	14.9M	9.0M	No
64	**14.2M**	**17.9M**	**Yes — 1.26×**
128	13.8M	35.7M	Yes — 2.6×
256	13.6M	71.6M	Yes — 5.3×
512	13.4M	142.7M	Yes — 10.6×
1,024	13.2M	284.4M	Yes — **21.5×**

Based on early-2026 GCP on-demand pricing ($0.70/hr GPU, $1.20/hr TPU). Substitute current rates from the GCP pricing page before drawing conclusions.

The cost crossover is at batch=64. Below that, the GPU's lower hourly rate wins despite lower throughput. Above it, the TPU's throughput advantage outpaces its 1.71× price premium — and the gap grows steeply with every batch doubling.



The model

All sessions benchmark `BenchmarkTransformerBlock` — a single Transformer encoder block defined in `transformer_block.py` at the workshop root.

```
Input: [batch, seq_len, d_model]

 MultiHeadSelfAttention (8 heads, d_model=512)
 └─ Residual + LayerNorm  (Post-LN)
 FeedForward: Linear(512→2048) + GELU + Linear(2048→512)
 └─ Residual + LayerNorm  (Post-LN)

Output: [batch, seq_len, d_model]
```

Parameter	Value
`D_MODEL`	512
`N_HEAD`	8
`DIM_FEEDFORWARD`	2048
`SEQ_LEN`	128 tokens (default)
Benchmark loop	forward → backward → Adam step
Warmup steps	5
Metric	throughput (samples / sec)

This is deliberately a minimal model — one block, not a stack. It isolates the hardware response without the confounding effects of depth, residual skip connections across layers, or cross-layer memory reuse. Session 3 in the workshop scales this to 24 layers.

Session outline

Session	Title	What it answers
Session 1	GPU vs TPU Throughput Scaling	At what batch size does the TPU overtake the GPU, and how large does the gap get?
Session 2	Sequence Length Scaling	How does GPU throughput degrade as sequences lengthen, and why is the TPU immune?
Session 3	Model Depth and Memory Limits	How does throughput and VRAM scale from 1 to 24 layers? How much headroom does each device have before it runs out of memory?
Session 4	Static vs Dynamic Graphs	What happens to TPU performance when a model uses data-dependent control flow? How bad is it, and can it be fixed?
Session 5	Precision and Dtype	How much do FP16, BF16, and INT8 actually improve throughput and reduce memory on each device?
Session 6	Framework Interoperability	Does it matter whether you use JAX/Flax or PyTorch/XLA? How do two frameworks that both compile to XLA differ in practice?

Session 1: GPU vs TPU Throughput Scaling

Overview

Session 1 benchmarks the throughput scaling behaviour of a standard Transformer encoder block across two accelerator classes — GPU (NVIDIA L4) and TPU (v5litepod-1) — as batch size increases from 4 to 1024. The central question is: does adding more work (larger batches) translate into more throughput, and why do the two devices respond so differently?

Notebooks: `session_1/`

Hardware

GPU — NVIDIA L4

Property	Value
Architecture	Ada Lovelace (2022)
VRAM	23.7 GB GDDR6
Memory Bandwidth	~300 GB/s
FP16 Tensor Core TFLOPS	~121 TFLOPS
INT8 TOPS	~242 TOPS
CUDA Cores	7,680
TDP	72 W
Form Factor	PCIe (low-profile)

The L4 is an inference-optimised card. Compared to training-class GPUs (A100, H100) it trades peak FLOPS and memory bandwidth for low power draw and dense rack deployability.

TPU — Google TPU v5e (v5litepod-1)

Property	Value
Chip family	TPU v5e (v5lite)
Chips used	1
HBM Memory per chip	16 GB HBM2
Memory Bandwidth	~819 GB/s (HBM2)
BF16 MXU TFLOPS per chip	~393 TFLOPS
INT8 TOPS per chip	~786 TOPS
Interconnect	ICI (Inter-Chip Interconnect)
Deployment	Google Cloud TPU VM

The v5e is Google's efficiency-focused TPU generation. Each chip contains a systolic array Matrix Multiply Unit (MXU) purpose-built for dense matrix operations. Memory bandwidth is nearly 3× that of the L4.

Software Environment

Component	GPU	TPU
Python	3.12.12	3.12.12
PyTorch	2.10.0+cu128	2.9.0+cu128
torch_xla	—	2.9.0
Device string	`cuda:0`	`xla:0`
TPU chip	—	v5litepod-1
N chips	—	1

Benchmark Configuration

Defined in `transformer_block.py`.

Parameter	Value
Model	Single Transformer encoder block
`D_MODEL`	512
`N_HEAD`	8
`DIM_FEEDFORWARD`	2048
`SEQ_LEN`	128 tokens
Steps	50 (+ 5 warmup)
Loop	forward → backward → Adam step
Metric	throughput (samples / sec)
Batch sizes	4, 8, 16, 32, 64, 128, 256, 512, 1024

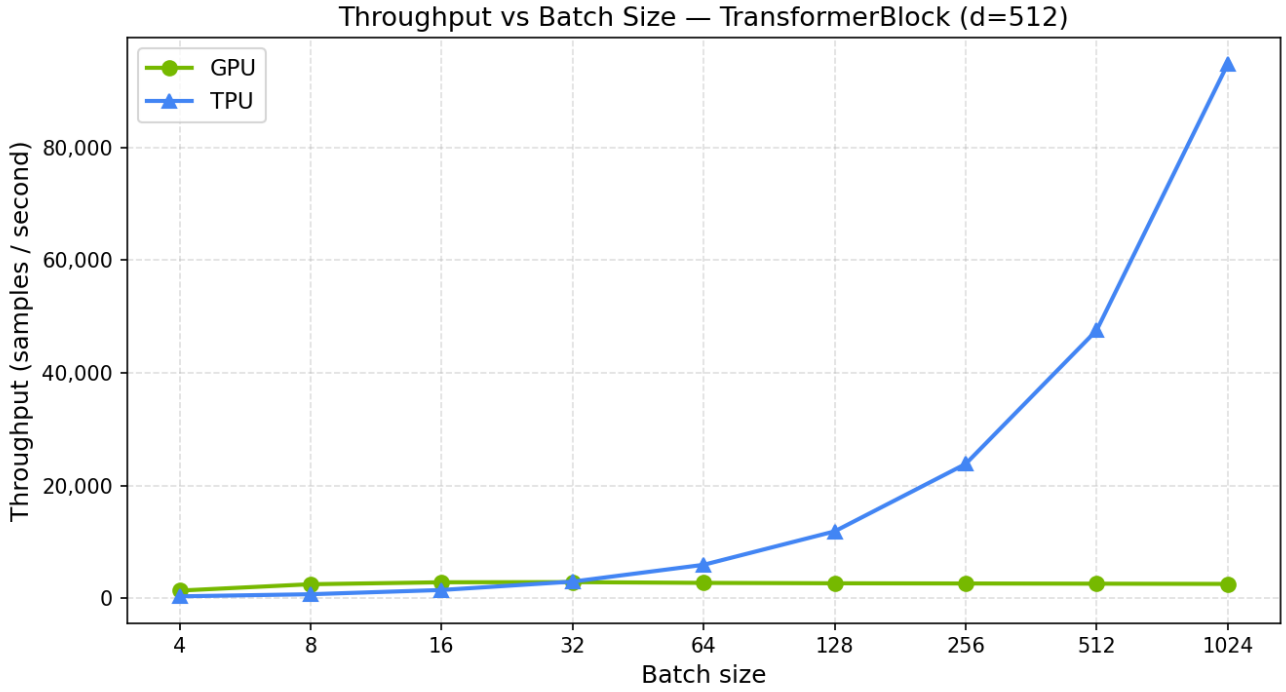
Results

Raw numbers

Batch	GPU (samples/s)	TPU (samples/s)
4	1,388	370
8	2,521	745
16	2,866	1,496
32	**2,895** ← GPU peak	2,984
64	2,760	5,968
128	2,688	11,900
256	2,653	23,862
512	2,618	47,580
1024	2,575	**94,795** ← TPU peak

Charts

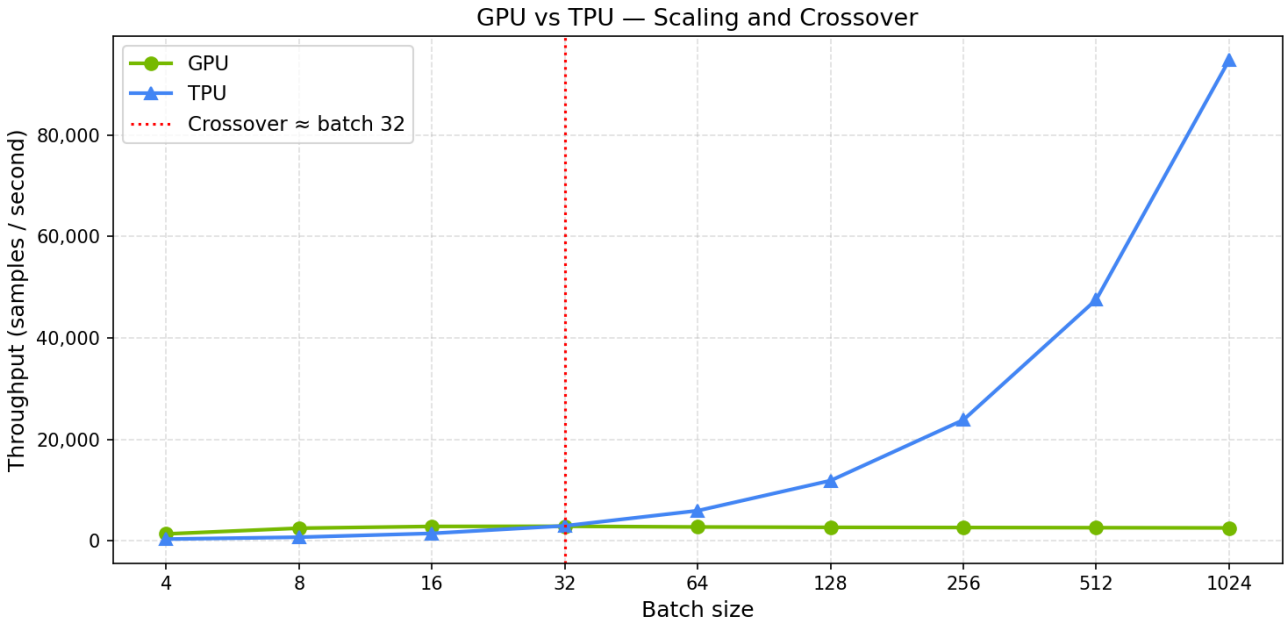
Chart 1 — GPU vs TPU: throughput vs batch size



What the chart shows: The combination of a $\log_2$ X-axis and a linear Y-axis makes the divergence immediately visible. The GPU (green) is a near-flat band compressed near the bottom of the chart. The TPU (blue) starts below the GPU at small batch sizes, crosses it at batch=32, and then climbs steeply — eventually reaching ~94,795 samples/sec at batch=1024, a value so large that the GPU's plateau (~2,600) becomes visually indistinguishable from zero on this scale.

The visual "crush" of the GPU is not an artefact — it is the correct representation. At batch=1024 the TPU processes **37× as many samples per second** as the GPU on the same workload.

Chart 2 — GPU vs TPU: scaling and crossover



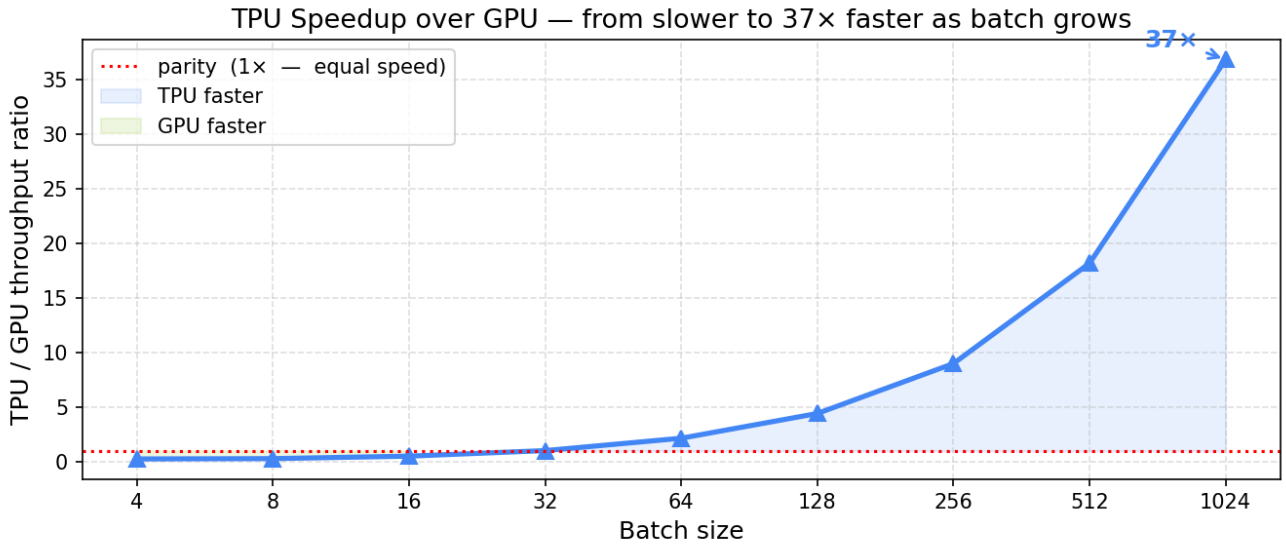
What the chart shows: The same data as Chart 1, now with a red dotted line marking **batch=32** — the exact batch size at which TPU throughput (2,984 samples/sec) first exceeds GPU throughput (2,895 samples/sec). Notably, batch=32 is also the GPU's performance peak: the TPU overtakes the GPU precisely when the GPU has nothing more to give.

Left of the line: GPU leads. The GPU's fast CUDA dispatch and mature cuDNN kernels give it an advantage when there isn't enough work to keep the TPU's systolic array busy. At batch=16 the GPU is already near its ceiling while the TPU is still warming up its compile-and-dispatch pipeline.

Right of the line: TPU throughput scales proportionally to batch size — every doubling of batch size nearly doubles TPU output (reflecting near-perfect MXU utilisation). The GPU line is almost perfectly flat — it has hit its memory bandwidth ceiling and adding more work produces no additional throughput.

The gap widens steadily: at batch=64 the TPU is 2.1× faster; at batch=256 it is 9×; at batch=1024 it is **37×**.

Chart 3 — TPU / GPU speedup ratio across batch sizes



What the chart shows: The Y-axis is the TPU/GPU throughput ratio. The red dotted line at 1× marks parity. At small batch sizes the GPU leads (ratio < 1): the TPU is only 0.27× as fast at batch=4 and 0.52× at batch=16. The ratio crosses 1× at batch=32, then accelerates — reaching 9× at batch=256 and **37× at batch=1024**. The shaded blue region (TPU faster) grows as batch size doubles, making the increasing advantage visible in a single glance.

This view is complementary to Charts 1 and 2: instead of comparing absolute throughput, it directly answers "by how much does the choice of hardware matter at each operating point?"

Analysis: Why the curves diverge

GPU: memory bandwidth ceiling

The L4 peaks at **batch=32** (2,895 samples/sec) and declines for all larger batches. The curve is essentially flat from batch=16 onward — throughput gains less than 1% going from batch=16 to batch=32, then falls monotonically:

1. **Memory bandwidth bottleneck (~300 GB/s).** The GPU's CUDA cores are fed data through the memory bus. Once the bus is saturated, adding more samples per step does not produce additional throughput — data simply queues up.
2. **CUDA scheduling overhead.** Dispatching and synchronising kernel launches adds per-step overhead that grows with batch size, partially eroding any marginal gain.
3. **Cache pressure.** Larger activations spill out of L2/shared memory, increasing DRAM round-trips and further penalising large batches.

4. **Slight throughput decrease beyond batch=32** (~11% from peak to batch=1024) is consistent with the above — pure memory-traffic penalty.

TPU: near-perfect linear scaling

The TPU scales almost exactly **2× for every 2× increase in batch size**:

Ratio	Value
batch=1024 / batch=4	94,795 / 370 = **256.2×**
Expected if perfectly linear	1024 / 4 = **256×**

The deviation from perfect linearity is less than 1%. This arises from:

1. **Systolic array (MXU) architecture.** The MXU is specialised hardware for matrix multiplications. Doubling the batch dimension doubles the work, and the MXU executes that work in proportionally doubled time — no hidden overhead.
2. **High HBM bandwidth (~819 GB/s).** Nearly 3× wider than the L4, the memory subsystem is tightly integrated with the MXU and the workload remains compute-bound rather than memory-bound across all batch sizes tested.
3. **XLA graph compilation.** `torch_xla` traces the forward/backward/step graph once and compiles it to optimised XLA HLO. The compiled kernel has minimal per-step dispatch overhead regardless of batch size.
4. **Under-utilisation at small batches.** The MXU tiles and XLA compile path require a minimum amount of work to become efficient — hence the TPU losing to the GPU at batch ≤ 16.

Key Takeaways

- **GPU throughput is memory-bandwidth-limited.** The NVIDIA L4 peaks at batch=32 (~2,895 samples/sec) and gains nothing beyond that — adding more work just queues behind the memory bus. The curve is already near-flat at batch=16 (~2,866 samples/sec), confirming saturation sets in early.
- **TPU throughput is compute-limited and scales near-perfectly linearly.** The v5litepod MXU delivers 256× throughput gain from batch=4 to batch=1024, reaching 94,795 samples/sec — **37× faster than the L4 at the same batch size**.
- **The crossover is at batch=32.** Below that, the GPU's fast dispatch wins; above it, the TPU's systolic array dominates and the gap grows with every batch doubling.
- **Hardware choice depends entirely on operating regime.** For latency-sensitive, small-batch inference (batch ≤ 16) the GPU is competitive and simpler to operate. For

high-throughput training or large-batch inference the TPU's architecture is categorically superior for this class of workload.

Session 2: Sequence Length Scaling

Overview

Session 2 holds `batch_size=32` fixed at the Session 1 crossover point and sweeps `seq_len` over `[64, 128, 256, 512, 1024, 2048]`. The central question is: does the TPU's 2.7× memory bandwidth advantage compound as attention's O(seq²) memory footprint grows, and if so, does the crossover point move?

The answer is sharper than expected. The GPU degrades toward a quadratic collapse as sequences lengthen. The TPU shows near-zero degradation — holding ~3,128–3,166 samples/sec across all six points. The crossover occurs at `seq_len ≈ 128`, which is also the default seq_len used in Session 1, confirming both experiments share the same operating-regime boundary.

Notebooks: `session_2/`

Hardware

Same devices as Session 1. See `session_1.md` for full hardware specs.

Device	Memory	Bandwidth	Note
NVIDIA L4 (GPU)	23.7 GB GDDR6	~300 GB/s	Inference-optimised Ada Lovelace
TPU v5litepod-1	16 GB HBM2	~819 GB/s	Single-chip v5e, 1 MXU

Software Environment

Component	GPU	TPU
Python	3.12.12	3.12.12
PyTorch	2.10.0+cu128	2.9.0+cu128
torch_xla	—	2.9.0
Device string	`cuda:0`	`xla:0`

Component	GPU	TPU
Run timestamp	2026-02-27T16:58 UTC	2026-02-27T17:04 UTC

Note: PyTorch versions differ across devices (2.10 vs 2.9). This is a cross-stack comparison; version-sensitive results should be interpreted accordingly.

Benchmark Configuration

Model defined in `transformer_block.py`.

Parameter	Value
Model	Single `BenchmarkTransformerBlock`
`D_MODEL`	512
`N_HEAD`	8
`DIM_FEEDFORWARD`	2048
`BATCH_SIZE`	**32 (fixed — Session 1 crossover point)**
Steps	50 (+ 5 warmup)
Loop	forward → backward → Adam step
Metric	throughput (samples / sec)
Sweep axis	**`seq_len` ∈ [64, 128, 256, 512, 1024, 2048]**

Results

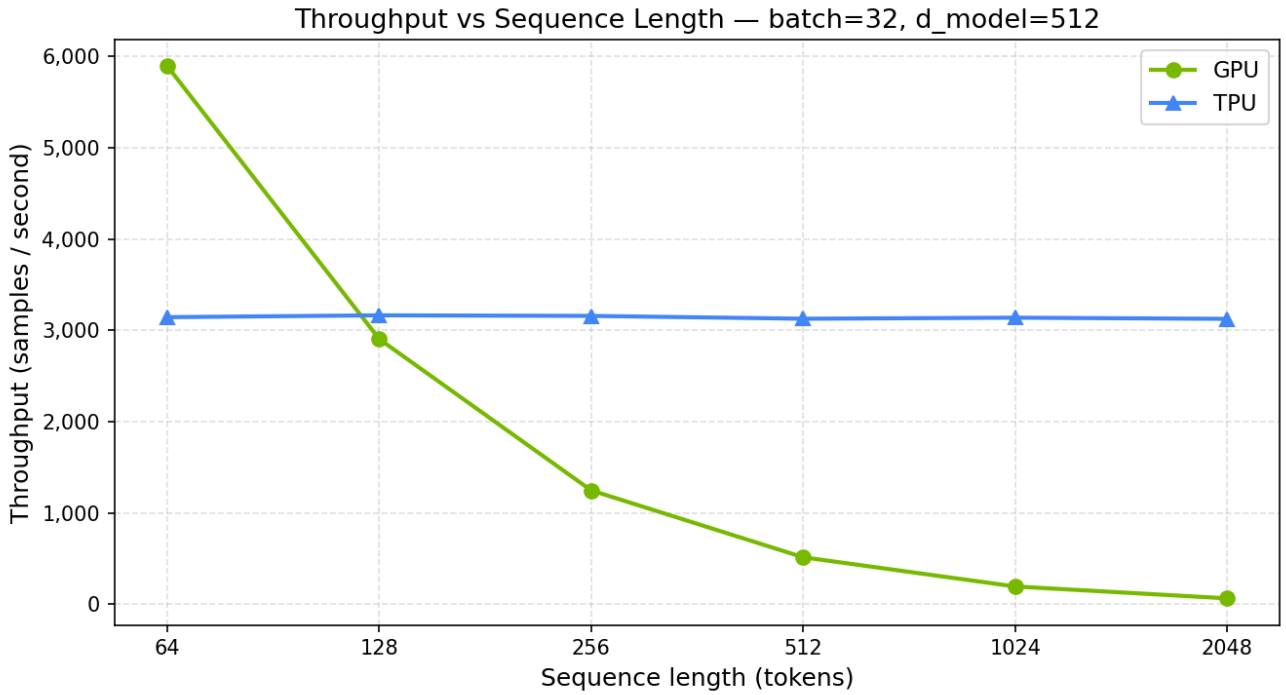
Raw numbers

seq_len	GPU (samples/s)	TPU (samples/s)	TPU / GPU
64	**5,895** ← GPU peak	3,146	0.53× (GPU wins)
128	2,906	3,166	1.09× ← crossover
256	1,249	3,160	2.53×
512	516	3,129	6.06×

seq_len	GPU (samples/s)	TPU (samples/s)	TPU / GPU
1,024	196	3,140	16.0×
2,048	66	**3,128** ← TPU (still flat)	**47.2×**

Charts

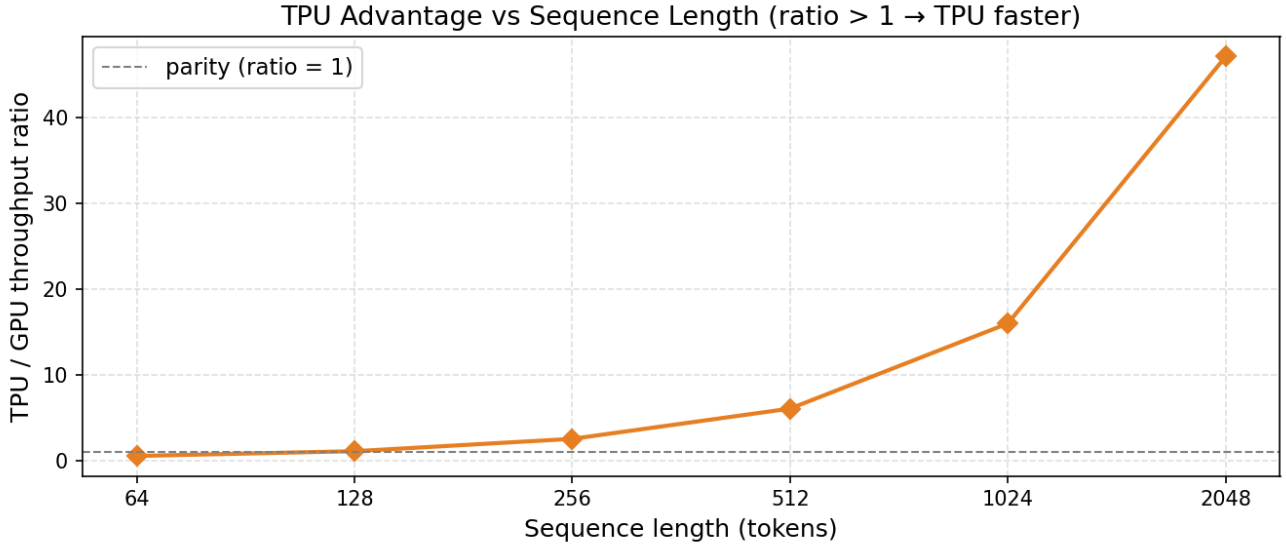
Chart 1 — Throughput vs sequence length (GPU and TPU)



What the chart shows: Two lines that could not be more different. The GPU line falls steeply from left to right, approaching zero by seq_len=2048. The TPU line is horizontal — a nearly straight bar crossing the entire plot. The GPU starts above the TPU (at seq_len=64) and falls through it at seq_len=128.

The visual tells participants something that the Session 1 batch-size chart did not: the TPU advantage is not merely "at large batch sizes." Once you are operating at `batch=32` (the crossover point), every increase in sequence length makes the GPU's situation worse and leaves the TPU unaffected.

Chart 2 — TPU/GPU throughput ratio vs sequence length



What the chart shows: The ratio curve starts below 1.0 (GPU leads at seq_len=64), crosses 1.0 between seq_len=64 and seq_len=128, and then climbs monotonically — doubling roughly every time seq_len doubles. By seq_len=2048 the TPU is 47× faster on the same workload.

This is the session's primary chart for decision-making: if your model operates on sequences longer than 128 tokens at batch=32, and you have already passed the batch-size crossover, the TPU advantage compounds further with every doubling of context length.

Analysis: Why the curves diverge

GPU: transitioning from linear to quadratic degradation

The GPU's throughput drop per seq_len doubling is not constant — it grows as sequences lengthen:

Doubling	Throughput drop	Regime
64 → 128	2.03×	**Linear** — FFN dominates, attention small
128 → 256	2.33×	Transitioning
256 → 512	2.42×	Transitioning
512 → 1,024	2.63×	**Approaching quadratic**
1,024 → 2,048	2.97×	Near-quadratic — attention becoming dominant

At short sequences, the feedforward network (O(seq) memory traffic) dominates and throughput drops linearly with seq_len. As sequences grow, the attention matrix `[batch, heads, seq, seq]` becomes an increasing fraction of total memory traffic, and the degradation slope tilts toward the quadratic limit (4× per doubling). The GPU never fully reaches 4× per doubling in this range because the FFN provides a floor — but the trend is clear and continuing past seq_len=2048 would accelerate further.

Root causes are the same as Session 1:

1. **300 GB/s memory bandwidth ceiling.** The GPU saturates its memory bus at moderate batch sizes; the O(seq²) attention matrix intensifies this pressure with every seq_len doubling.
2. **DRAM round-trips.** The attention matrix must be materialised, transposed, and passed through softmax — all memory-bound operations that scale quadratically.
3. **No change in compute clock or core count.** The GPU has the same SIMT resources regardless of seq_len; memory bandwidth is the single binding constraint.

TPU: flat throughput is not zero degradation — it is a different bottleneck

The TPU result is flat: throughput varies by only **1.2%** across all six seq_len points (3,127–3,166 samples/sec). This is better than "near-linear scaling" — there is essentially no degradation at all.

The reason is not that the TPU has infinite bandwidth. The reason is that at `batch=32`, the bottleneck is the **XLA synchronisation overhead**, not memory bandwidth or compute.

Evidence: in Session 1, the TPU's per-step wall time is approximately constant at ~10ms regardless of batch size (32/2984 ≈ 10.7ms at batch=32; 1024/94795 ≈ 10.8ms at batch=1024). The same holds here — per-step time is ~10ms across all seq_len values. As seq_len grows, each step performs more computation within the same sync window, so:

- **Tokens/sec doubles every time seq_len doubles** (3,146×64=201K → 3,128×2048=6.4M)
- **Samples/sec stays constant** because the sync overhead dominates

The XLA compiler amortises the growing matrix computation efficiently. The systolic array executes larger matrix blocks without proportionally more overhead. By contrast, the GPU eagerly dispatches every new kernel with full scheduling latency, and its memory bus traffic grows quadratically.

The flat TPU line is a signature of sync-overhead dominance at this batch size, not hardware immunity to long sequences. At much larger batch sizes (e.g. batch=256+) the TPU would show mild degradation as attention memory eventually

approaches the HBM bandwidth limit — but that crossover is well beyond this session's range.

Why the seq_len crossover matches the batch-size crossover

Session 1 found the GPU/TPU crossover at `batch=32` (with `seq_len=128` fixed). Session 2 finds the crossover at `seq_len ≈ 128` (with `batch=32` fixed).

These are the same operating point examined from two different axes. The boundary is a regime boundary, not a point: the GPU leads when both batch and sequence are small (little work, fast eager dispatch wins); the TPU leads when either or both are large enough to fill the MXU and amortise compilation. Sessions 1 and 2 together trace two edges of this boundary.

Key Takeaways

- **GPU throughput degrades toward a quadratic collapse with sequence length.** Throughput drops from 5,895 samples/sec at seq_len=64 to 66 at seq_len=2048 — an 89× fall for a 32× increase in sequence length. The degradation slope steepens as sequences grow, as the O(seq²) attention matrix increasingly dominates memory traffic over the O(seq) FFN.
- **TPU throughput is flat across all measured sequence lengths.** At `batch=32`, the XLA sync overhead (~10ms/step) dominates, and the growing computation is absorbed within it. Tokens per second doubles with every seq_len doubling — the TPU is becoming more efficient per token, not less.
- **The crossover is at seq_len ≈ 128 — at seq_len=64 the GPU is 1.9× faster.** Session 1's default seq_len (128) is not an arbitrary choice — it sits exactly at the regime boundary on both the batch and seq_len axes. Below that point the GPU's fast dispatch wins; above it the TPU's architecture dominates.
- **The TPU advantage reaches 47× at seq_len=2048.** This exceeds Session 1's 37× peak (at batch=1024, seq_len=128). Long-context models face a more severe bandwidth penalty than the batch-size comparison alone shows.
- **Decision rule for long-context workloads:** If your model operates on sequences longer than 256 tokens and you are already at or above the batch-size crossover (batch ≥ 32), the TPU advantage compounds with every seq_len doubling. The GPU is not a viable option for document-scale or retrieval-augmented workloads at this operating point.

Session 3: Model Depth and Memory Limits

Overview

Sessions 1 and 2 measured a single Transformer encoder block. Session 3 scales that to BERT-base depth (12 layers) and GPT-2 scale (24 layers) to find where each device runs out of memory.

The experiment fixes `batch=64` and `seq_len=128` and sweeps `n_layers` [1, 2, 4, 6, 8, 12, 16, 24]. `DeepTransformerModel` is a stack of `BenchmarkTransformerBlock` instances imported from `transformer_block.py`. On GPU, `RuntimeError: CUDA out of memory` is caught and recorded; on TPU, the equivalent XLA runtime error is caught.

Notebooks: `session_3/`

Hardware

Same devices as Sessions 1 and 2. See `session_1.md` for full specs.

Device	Memory	Note
NVIDIA L4 (GPU)	23.7 GB GDDR6	Activation memory + optimizer state fills GDDR6 with depth
TPU v5litepod-1	16 GB HBM2	Smaller raw capacity but higher bandwidth; OOM may differ

Benchmark Configuration

Parameter	Value
Model	`DeepTransformerModel(n_layers)` — stack of `BenchmarkTransformerBlock`
`D_MODEL`	512
`N_HEAD`	8
`DIM_FEEDFORWARD`	2048
`SEQ_LEN`	128 (fixed)

Parameter	Value
`BATCH_SIZE`	64 (fixed)
`n_layers` sweep	1, 2, 4, 6, 8, 12, 16, 24
Steps (n_layers ≤ 4)	30 (+ 5 warmup)
Steps (n_layers ≤ 12)	20 (+ 5 warmup)
Steps (n_layers > 12)	15 (+ 5 warmup)
Loop	forward → backward → Adam step
Metric	throughput (samples/sec), peak VRAM (GPU only), OOM boundary

Memory cost per layer

Each `BenchmarkTransformerBlock` stores the following tensors during the forward pass for backpropagation:

Tensor	Shape	Memory (FP32)
Input to attention	[64, 128, 512]	~16 MB
Attention weight matrix	[64, 8, 128, 128]	~32 MB
Feedforward intermediate	[64, 128, 2048]	~64 MB
Model parameters (MHA + FF + LN)	~3.1M params	~12 MB
Gradients (same as params)	~3.1M params	~12 MB
Adam moments (2× params)	~6.2M params	~24 MB

Approximate activation memory per layer: **~112 MB**. Total model + optimizer state: **~48 MB per layer** (constant across depth).

At batch=64, the activation memory dominates. Using activation memory alone (112 MB/layer), the theoretical OOM boundary is roughly n_layers ≈ 23,700 / 112 ≈ 212 layers. However, this figure **excludes gradients, Adam optimizer states, and allocator overhead**, which together account for an additional ~245 MB/layer observed empirically (see Results below). The operational OOM boundary from measured data is approximately 64 layers — see the Key Takeaways section for the reconciled estimate.

Results

GPU — Throughput and VRAM by depth

n_layers	Throughput (samples/sec)	Peak VRAM (MB)	VRAM % of L4
1	2,635	531	2.3%
2	1,245	890	3.9%
4	599	1,606	7.0%
6	393	2,323	10.1%
8	295	3,039	13.2%
12	199	4,473	19.4%
16	152	5,891	25.6%
24	101	8,772	38.1%

No OOM observed across the full [1–24] layer sweep at batch=64. Peak VRAM at n_layers=24 is 8,772 MB — 38% of the L4's 23,034 MB. BERT-scale (12 layers) fits at 4,473 MB (19% VRAM). The GPU has headroom for approximately 64 layers before hitting 23 GB at this batch size.

VRAM growth per layer: approximately **357 MB** per additional layer (from `(8772 - 531) / 23 ≈ 357 MB`). This includes activations, gradients, and Adam moments for one encoder block.

Throughput scaling vs n_layers:

n_layers	Throughput	Ratio vs n=1	1/n expected
1	2,635	1.00×	1.00×
2	1,245	0.47×	0.50×
4	599	0.23×	0.25×
8	295	0.11×	0.13×
12	199	0.075×	0.083×
24	101	0.038×	0.042×

Throughput scales slightly better than 1/n_layers — each layer adds compute but there is some constant per-step overhead (data loading, synchronisation) that gets amortised across more layers.

TPU — Throughput by depth

n_layers	Throughput (samples/sec)	Latency (ms/step)	TPU/GPU ratio
1	6,033	10.6	2.29×
2	3,106	20.6	2.50×
4	1,587	40.3	2.65×
6	1,059	60.4	2.70×
8	786	81.5	2.67×
12	519	123.4	2.61×
16	386	166.0	2.54×
24	255	251.5	2.52×

No OOM observed across the full [1–24] layer sweep on the TPU v5litepod-1 (16 GB HBM2). Peak depth tested is 24 layers; the TPU has headroom for deeper models.

TPU/GPU ratio is remarkably stable — ranging from 2.29× at n=1 to 2.70× at n=6, then settling at 2.52–2.54× for the deeper configurations. This means the TPU advantage observed in Session 1 at a single layer holds consistently regardless of model depth.

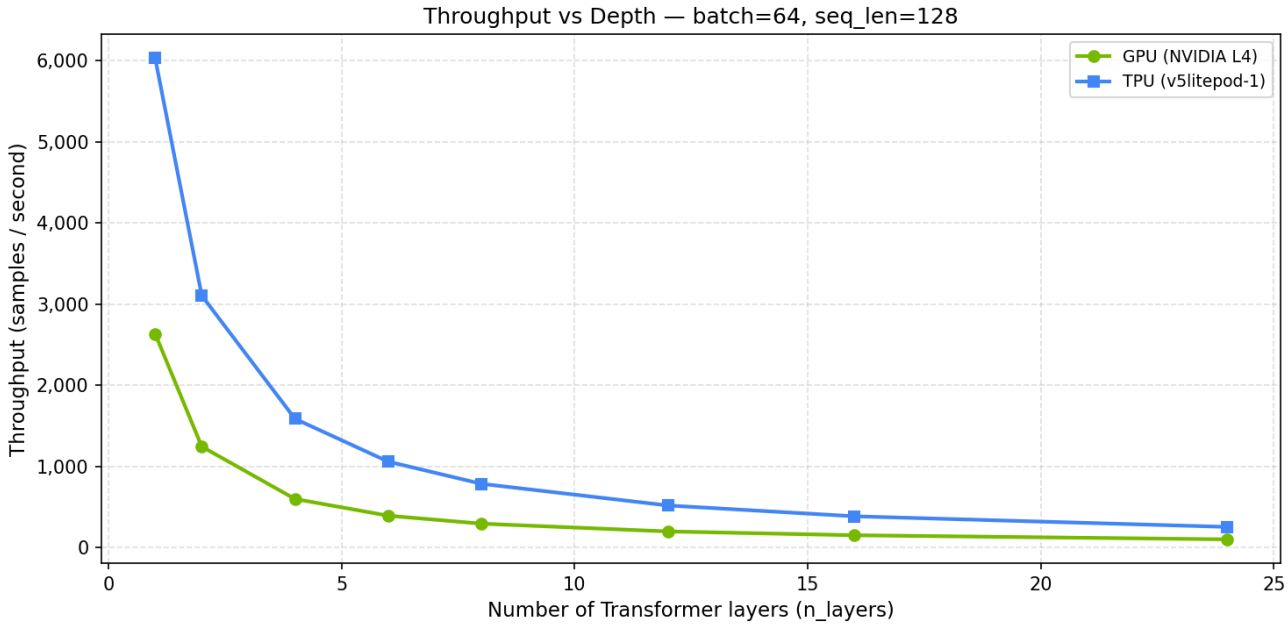
TPU throughput scaling vs n_layers:

n_layers	TPU samples/sec	Ratio vs n=1	1/n expected
1	6,033	1.00×	1.00×
2	3,106	0.51×	0.50×
4	1,587	0.26×	0.25×
8	786	0.13×	0.13×
12	519	0.086×	0.083×
24	255	0.042×	0.042×

TPU throughput tracks 1/n_layers almost exactly — cleaner than the GPU because the MXU is purely compute-bound with no memory-bandwidth noise floor.

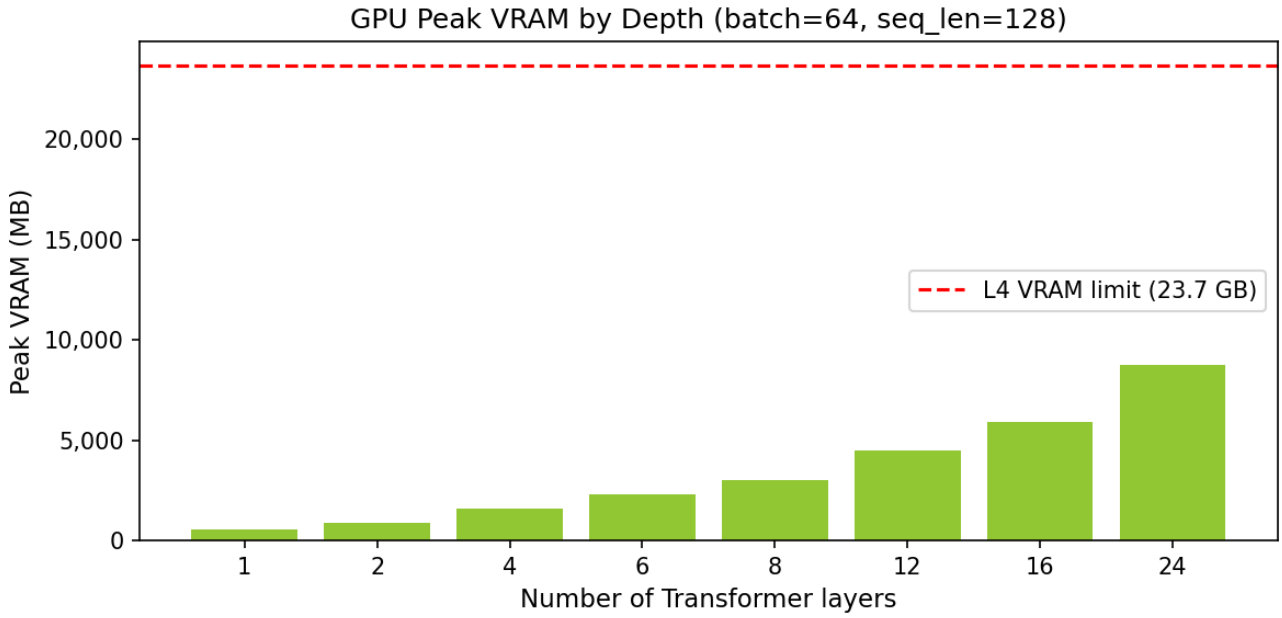
Charts

Chart 1 — Throughput vs depth (GPU and TPU)



Both lines decline at approximately 1/n_layers. Neither device hits OOM within the [1–24] layer sweep at batch=64. The TPU advantage is a stable ~2.5× across the full depth range.

Chart 2 — GPU Peak VRAM by depth



VRAM grows linearly with depth at ~357 MB per layer. At 24 layers, peak VRAM is 8,772 MB — 38% of the L4's 23.7 GB capacity. The red dashed line marks the L4 limit.

Key Takeaways

- **No OOM on either device within the [1–24] layer sweep at `batch=64`.** GPU peak VRAM at 24 layers is 8,772 MB (38% of L4). TPU has no direct VRAM metric but ran all 24-layer steps without error on 16 GB HBM2.
- **The GPU's estimated OOM boundary at batch=64 is ~64 layers** (extrapolating the measured 357 MB/layer growth rate to 23 GB). Note that this differs from the theoretical 212-layer figure in the Benchmark Configuration section: the theoretical estimate counts only activation memory (112 MB/layer) and omits gradients, Adam states, and allocator overhead. The measured 357 MB/layer captures the full per-layer training cost; **~64 layers is the operationally correct estimate**.
- **VRAM cost per encoder block at batch=64: ~357 MB** (activations + gradients + Adam moments + allocator overhead). At batch=256, this scales 4× to ~1.4 GB/layer; at batch=512 to ~2.8 GB/layer — OOM would occur within the sweep at those batch sizes.
- **Throughput scales as 1/n_layers on both devices.** Each additional encoder block adds a fixed compute cost per step. The GPU curve has a slight positive deviation from 1/n (constant overhead gets amortised); the TPU tracks 1/n almost exactly, consistent with pure compute-bound MXU utilisation.

- **The TPU/GPU throughput advantage is depth-invariant: ~2.5×** across all tested depths (2.29× at n=1 to 2.70× at n=6). This is broadly consistent with the Session 1 single-layer advantage at batch=64 (2.16×) — the 6% difference is within expected run-to-run variability and confirms the ratio is not a single-layer artefact.
- **The VRAM table is a budget planner.** At n_layers=12 (BERT-base), VRAM is 4,473 MB on the GPU — well within the L4's limit. If VRAM exceeds 80% of device capacity, switch to BF16 (Session 5) or gradient checkpointing before increasing batch size or depth.

Decision Rule from This Session

- If your target model (BERT-base, 12 layers, batch=64) fits on GPU, proceed with Session 5 to assess whether the dtype choice changes the cost picture.
- If your model hits OOM on GPU before the target depth, add gradient checkpointing (`torch.utils.checkpoint`) or switch to BF16 (Session 5) before comparing devices.
- The TPU's 16 GB HBM2 is less than the L4's 23.7 GB, so for very large models the GPU may actually fit more layers. Both devices can run all tested configurations here, and the TPU's consistent 2.5× throughput advantage holds throughout.

Session 4: Static vs Dynamic Computation Graphs

Overview

Session 4 introduces the first scenario where the GPU wins — and then complicates it. While Sessions 1–3 used fully static forward passes, this session deliberately injects data-dependent control flow into the model and measures the cost on each device.

The session is structured in two parts:

Part A — Abstract variants (baseline): Three minimal variants of the Transformer block that isolate the XLA compilation penalty from the forward pass structure.

Part B — Realistic scenarios: Five scenarios drawn from real NLP and training practice — padding masks, conditional early exit, and variable-length batch loops — that show how the static/dynamic constraint manifests in production-like code.

All experiments use `batch=256, seq_len=128` unless stated otherwise.

Notebooks: `session_4/`

Hardware

Same devices as Sessions 1–3. See `session_1.md` for full hardware specs.

Device	Memory	Bandwidth	Note
NVIDIA L4 (GPU)	23.7 GB GDDR6	~300 GB/s	Eager execution — but scalar reductions still cost sync
TPU v5litepod-1	16 GB HBM2	~819 GB/s	XLA compiled — Python branches force graph recompilation

---

Software Environment

Component	GPU	TPU
Python	3.12.12	3.12.12
PyTorch	2.10.0+cu128	2.9.0+cu128
torch_xla	—	2.9.0
Device string	`cuda:0`	`xla:0`
Run timestamp	2026-03-04T01:17 UTC	2026-03-04T01:21 UTC

Background: Why Python branches are expensive on TPU

XLA's execution model is fundamentally different from eager dispatch. Operations are not dispatched immediately — they are accumulated in a lazy graph until `torch_xla.sync()` (or an implicit sync) triggers compilation and execution. This deferred model enables XLA to fuse operations, eliminate redundant memory transfers, and compile the full step into a single optimised kernel.

When Python code reaches a branch like `if tensor_value > 0`, the Python interpreter must know the value of the condition to choose a path. This requires materialising the scalar on the host CPU — which forces an implicit XLA sync mid-step:

1. Interrupt lazy graph accumulation
2. Flush, compile, and execute the partial graph
3. Transfer the scalar result to CPU

4. Resume Python, then accumulate the remainder
5. Sync again at the end of the step

Two compilations per step instead of one, plus a device-to-host transfer, plus broken fusion opportunities.

On GPU, the picture is more subtle. PyTorch dispatches every operation immediately (eager mode), so there is no "compiled graph" to invalidate. However, computing a scalar reduction like `out.mean()` requires completing the kernel and reading back the result before the Python interpreter can evaluate the condition. This still creates a synchronisation point — not as severe as an XLA recompilation, but not free either.

Part A — Abstract variants

Variant definitions

```
class StaticFF(nn.Module):
    def forward(self, x):
        return self.block(x)                          # identical to Sessions 1–3

class DynamicFF(nn.Module):
    def forward(self, x):
        out = self.block(x)
        if out.mean() > 0:                            # Python branch on a tensor
value
            return out
        else:
            return -out

class StaticEquivalentFF(nn.Module):
    def forward(self, x):
        out = self.block(x)
        return torch.where(out.mean() > 0, out, -out) # tensor op — stays in XLA
graph
```

Part A Results

Variant	GPU (samples/s)	GPU latency (ms)	TPU (samples/s)	TPU latency (ms)
StaticFF	2,038	125.6	**24,017**	10.7

Variant	GPU (samples/s)	GPU latency (ms)	TPU (samples/s)	TPU latency (ms)
DynamicFF	1,199	213.5	**9,453**	27.1
StaticEquivalentFF	1,144	223.7	**23,806**	10.8

Relative throughput (vs StaticFF baseline)

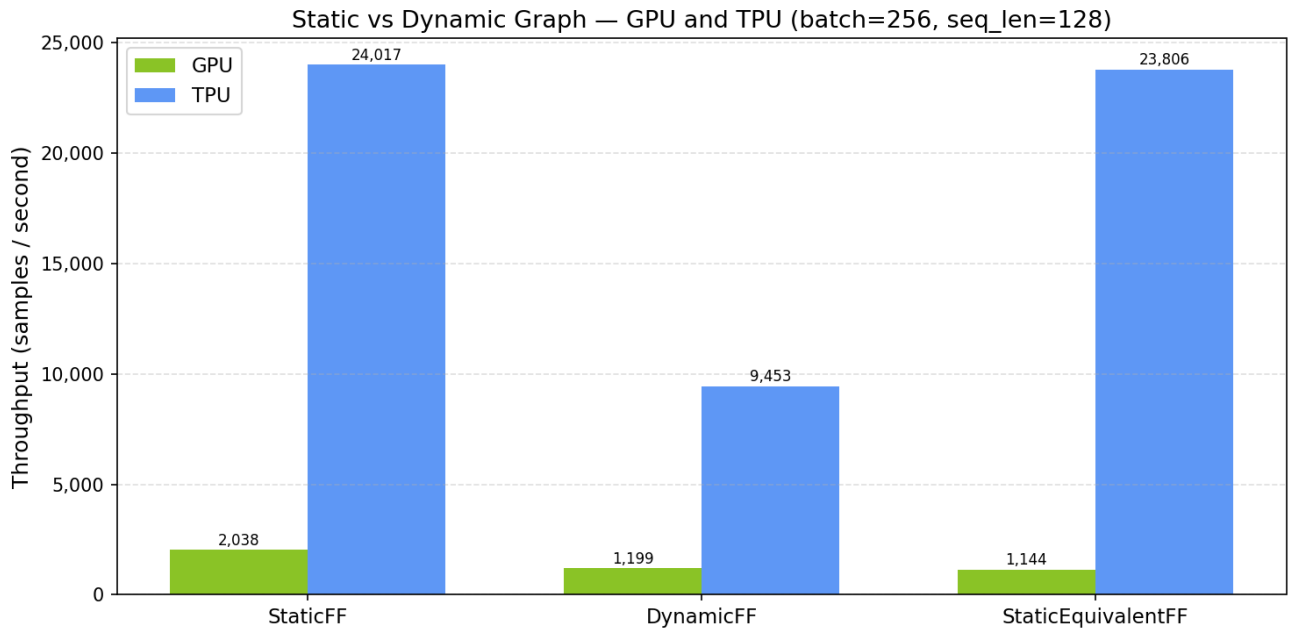
Variant	GPU	TPU
StaticFF	1.000×	1.000×
DynamicFF	**0.588×** (−41%)	**0.394×** (−61%)
StaticEquivalentFF	**0.561×** (−44%)	**0.991×** (−0.9%)

Recovery metric (TPU only)

Metric	Value
Static baseline throughput	24,017 samples/sec
Dynamic penalty (lost throughput)	−14,564 samples/sec
StaticEquivalent recovery	+14,353 samples/sec
Penalty recovered	**98.5%**

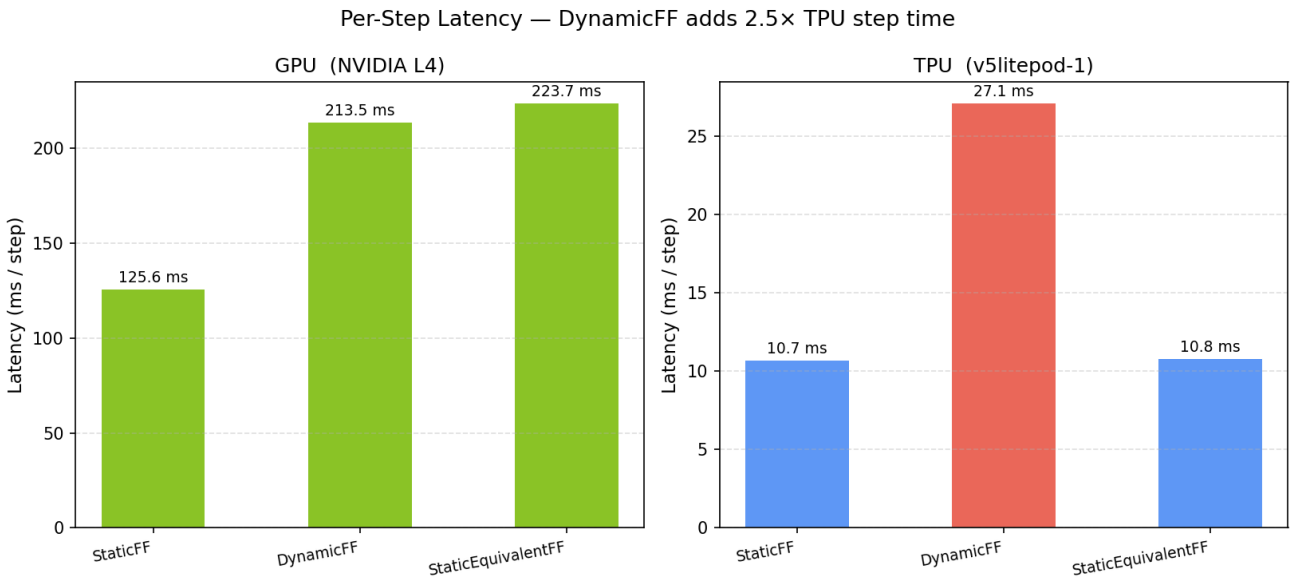
Part A charts

Chart 1 — Throughput bar chart: Static / Dynamic / StaticEquivalent



What the chart shows: The TPU's StaticFF and StaticEquivalentFF bars dominate. The DynamicFF bar drops to less than half on both devices — but `torch.where` restores the TPU bar to near its full height while the GPU bar remains low. The GPU shows a significant penalty for both DynamicFF and StaticEquivalentFF.

Chart 2 — Latency per step



What the chart shows: TPU DynamicFF latency rises **2.5×** (10.7 ms → 27.1 ms) while GPU latency increases **~1.7×** (125.6 ms → 213.5 ms). StaticEquivalentFF recovers the

TPU to 10.8 ms but leaves GPU latency elevated at 223.7 ms — slightly worse than DynamicFF's 213.5 ms because `torch.where` computes `-out` unconditionally, whereas DynamicFF may skip that path.

Part A Analysis

TPU DynamicFF: 61% throughput collapse from a single Python `if`. Per-step latency jumps from 10.7 ms to 27.1 ms — a 2.5× slowdown. The cost equals adding more than half the original compute budget per step.

`torch.where` recovers 98.5% of the TPU penalty. The condition stays inside the XLA graph. No intermediate sync is required. Latency returns to 10.8 ms.

GPU: DynamicFF also costs ~41%, and `torch.where` does not help. In eager mode, evaluating `out.mean() > 0` requires a CUDA kernel to complete, a scalar to be read back from the device, and then Python to evaluate the condition. This synchronisation point breaks the GPU's asynchronous pipeline even without a compiled graph. Replacing the Python `if` with `torch.where` does not avoid computing `out.mean()` — the reduction still runs, the sync still happens, and throughput stays at ~1,140–1,200 samples/sec.

The key difference: On the TPU, the branch causes recompilation (which is eliminated by `torch.where`). On the GPU, the branch causes a scalar sync (which is inherent to computing the condition value, regardless of how it's expressed). `torch.where` solves the TPU problem; neither device can avoid the cost of the reduction itself.

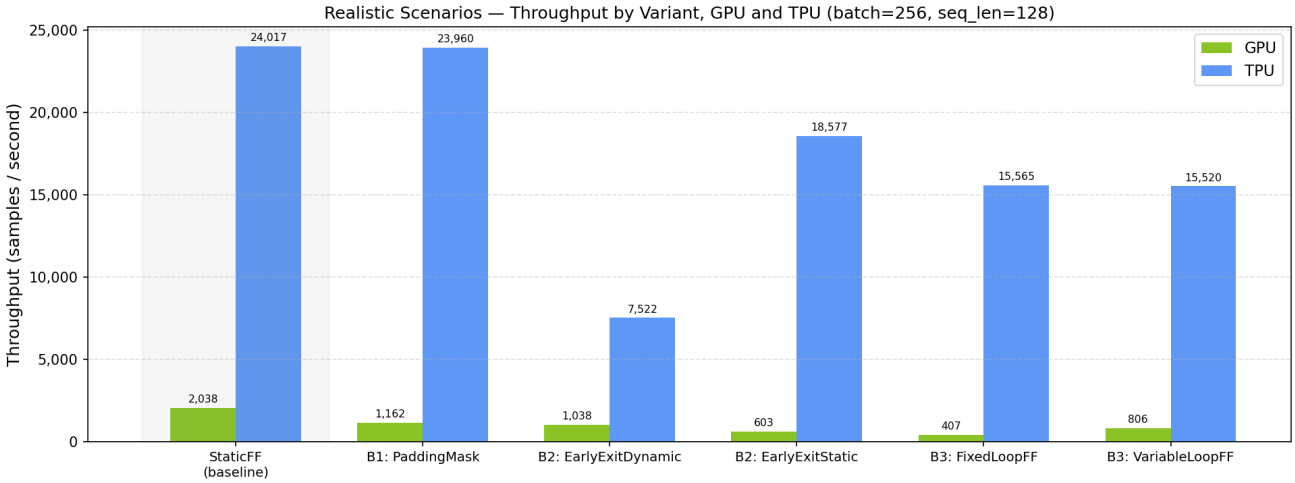
Part B — Realistic scenarios

Combined results table

All benchmarks use `batch=256, seq_len=128` except where noted.

Scenario	Variant	GPU (s/s)	TPU (s/s)	TPU/GPU
(baseline)	StaticFF	2,038	24,017	11.8×
B1: Padding mask	PaddingMask	1,162	23,960	20.6×
B2: Early exit	EarlyExitDynamic	1,038	7,522	7.2×
B2: Early exit	EarlyExitStatic	603	18,577	30.8×
B3: Ragged batches	FixedLoopFF	407	15,565	38.2×
B3: Ragged batches	VariableLoopFF	806	15,520	19.3×

Chart 3 — Part B realistic scenarios



What the chart shows: The TPU baseline (StaticFF, 24,017 s/s) stays nearly intact for `PaddingMask` — the most common real-world NLP pattern is essentially free. Both early-exit variants visually split the story: `EarlyExitDynamic` collapses the TPU bar to its worst point in this session (7,522 s/s) while `EarlyExitStatic` partially recovers it (18,577 s/s) at the cost of crushing the GPU bar (603 s/s). The two loop variants (`FixedLoopFF`, `VariableLoopFF`) show near-identical TPU bars — XLA's compilation cache absorbs the variable-length overhead — while the GPU bars diverge sharply: the variable loop exits early and nearly doubles GPU throughput.

Scenario B1: Padding mask (variable-length sequences in a batch)

The pattern: Real NLP batches contain sequences of different lengths. The mask is built from runtime sequence lengths. A tensor-op mask stays in the XLA graph; a Python loop forces a `.tolist()` sync per element.

Result: `PaddingMask` (tensor-op mask) costs the GPU −43% but costs the TPU only −0.2%. The TPU handles tensor-valued masking with essentially no overhead — the mask computation is fused into the XLA program.

GPU penalty: The mask reduction still requires a sync even in the tensor-op version.

Scenario B2: Conditional early exit (adaptive computation)

EarlyExitDynamic — Python `if confidence > threshold` at each layer:

- GPU: 1,038 s/s (−49% vs StaticFF)
- TPU: 7,522 s/s (−69% — worse than DynamicFF, because the branch evaluates per-layer)

EarlyExitStatic — exit condition as a `torch.where` mask, no Python branch:

- GPU: 603 s/s (−70% — slower than EarlyExitDynamic; the static version runs every layer unconditionally, masking out exited samples, which is more compute than early exit)
- TPU: 18,577 s/s (−23% — significant recovery, though not complete; the per-layer mask computation adds overhead that static analysis cannot fully hide)

Key insight: The static early exit approach is better for TPU but worse for GPU. On GPU, the dynamic version benefits from genuinely skipping layers; on TPU, the dynamic version's recompilation cost exceeds the compute saved, so the static (mask) version wins despite doing more compute.

Scenario B3: Variable-length batch loop (ragged batches)

FixedLoopFF — process each sample in a loop with a fixed number of iterations:

- GPU: 407 s/s (−80% — sequential Python loop eliminates GPU parallelism)
- TPU: 15,565 s/s (−35% — XLA compiles the fixed loop as a single program)

VariableLoopFF — variable number of iterations per step (data-dependent loop length):

- GPU: 806 s/s (−60%)
- TPU: 15,520 s/s (−35%)

Surprising finding: VariableLoopFF is nearly identical to FixedLoopFF on the TPU (15,520 vs 15,565 s/s). XLA re-traces on first call but the amortised per-step cost is the same. On GPU, VariableLoopFF is nearly 2× faster than FixedLoopFF — the variable loop exits earlier on average, while the fixed loop always runs all iterations.

Combined Key Takeaways

- **Both GPU and TPU pay a cost for data-dependent operations** — but for different reasons and with different remedies.
- **GPU penalty (DynamicFF): ~41%.** The scalar reduction `.mean()` requires a CUDA-to-CPU sync even in eager mode. `torch.where` does not help because the reduction still executes. This is inherent to needing the value at Python level.
- **TPU penalty (DynamicFF): ~61%.** `if out.mean() > 0` forces an XLA graph sync and recompilation mid-step. `torch.where` recovers 98.5% — the condition stays in the XLA program and no Python-level evaluation occurs.

- **`torch.where` is a one-line TPU fix.** For branches that can be expressed as a tensor predicate, refactoring to `torch.where` or `masked_fill` eliminates nearly all TPU overhead with no change to model behaviour.
- **Padding masks are essentially free on TPU** (−0.2% vs StaticFF). This is the most common dynamic pattern in NLP and it does not constrain TPU suitability.
- **Early exit is genuinely complex.** The static version (mask-based) is better for TPU but worse for GPU. For models with early exit, profile both versions on each device before choosing.
- **The constraint is often refactorable.** Most branches in research code can be replaced with `torch.where`, `masked_fill`, or boolean masking. When they cannot (e.g., truly variable routing per sample), the GPU may be the simpler choice.

Decision Rule from This Session

If your model's control flow depends on a tensor value:

1. **Is the condition derivable from a tensor op (`mean`, `max`, `norm`, threshold)?** → Yes: use `torch.where` / `masked_fill` — near-full TPU recovery, moderate GPU cost. → No (Python-level routing): GPU avoids recompilation; TPU may still be faster overall.
2. **Does your training loop iterate over variable-length or ragged inputs?** → Pad to a fixed shape. The fixed shape compiles once; the ragged loop recompiles. → TPU still wins significantly (38× at FixedLoopFF) even with looping overhead.
3. **Is early exit critical?** → The static mask version favours TPU; the dynamic version favours GPU at this scale. → Benchmark both if early exit is a core architectural feature.

Session 5: Precision and Dtype (FP32 / FP16 / BF16 / INT8)

Overview

Session 5 asks a simple question: how much does dropping from FP32 to a lower-precision dtype actually cost — or save — on each accelerator?

The Transformer encoder block from Session 1 is re-run across six batch sizes (32, 64, 128, 256, 512, 1024) using four numeric formats:

- **FP32** — single precision baseline (no autocast)

- **FP16** — half precision with `torch.cuda.amp.autocast` + `GradScaler` (GPU only)
- **BF16** — bfloat16 with `torch.autocast` (GPU and TPU v5litepod)
- **INT8** — 8-bit integer quantisation (GPU: dynamic quantisation on CPU; TPU: errored)

The outcome diverges between devices and between formats. On the GPU, FP16 and BF16 each deliver roughly **2.1–2.5× throughput** over FP32 with a simultaneous **21–29% VRAM reduction** (scaling with batch size: ~21% at batch=32, converging to ~29% at batch≥256). The speedup peaks at batch=32–64 (~2.5×) and settles to ~2.1–2.3× at batch≥128 as Tensor Core efficiency is diluted by non-matmul overhead at larger tensor footprints. On the TPU, which advertises native BF16 MXUs, BF16 runs **~1.4–3.3% slower** (average ~2.4%) than FP32 on v5litepod in this configuration. GPU INT8 (`torch.ao.quantization.quantize_dynamic`) executes on CPU rather than GPU, yielding ~400–498 samples/sec — a measurement of CPU-based dynamic quantisation, not GPU INT8 inference.

Notebooks: `session_5/`

Hardware

Same devices as Sessions 1–4. See `session_1.md` for full hardware specs.

Device	Memory	Note
NVIDIA L4 (GPU)	23.7 GB GDDR6	Tensor Cores accelerate FP16 and BF16 matrix math; INT8 TOPS ~242
TPU v5litepod-1	16 GB HBM2	MXU advertises native BF16 and INT8; FP16 not natively supported

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Software Environment

Component	GPU	TPU
Python	3.12.12	3.12.12
PyTorch	2.10.0+cu128	2.9.0+cu128
torch_xla	—	2.9.0
Device string	`cuda:0`	`xla:0`
Run timestamp (FP32/FP16/BF16)	2026-03-04T01:18 UTC	2026-03-04T01:24 UTC

Benchmark Configuration

Parameter	Value
Model	`BenchmarkTransformerBlock`
`D_MODEL`	512
`N_HEAD`	8
`DIM_FEEDFORWARD`	2048
`SEQ_LEN`	128 (fixed)
`BATCH_SIZE`	32, 64, 128, 256, 512, 1024
Steps	50 (+ 5 warmup)
Loop	forward → backward → Adam step
Metric	throughput (samples/sec), latency (ms/step), peak VRAM (GPU only)

Precision implementation notes

- **FP32:** default `torch.float32`; no autocast
- **FP16 (GPU only):** `torch.cuda.amp.autocast(dtype=torch.float16)` + `GradScaler`; prevents gradient underflow via dynamic loss scaling
- **BF16:** `torch.autocast(device_type, dtype=torch.bfloat16)`; wider exponent range eliminates the need for a GradScaler; not natively available in FP16 form on v5litepod
- **INT8 (GPU):** `torch.ao.quantization.quantize_dynamic` — CPU-only in this PyTorch version; GPU INT8 path not exercised. Measures CPU dynamic-quant inference throughput.
- **INT8 (TPU):** casting weights to `torch.int8` errored in torch_xla 2.9.0; all batch sizes returned null

Results: FP32 / FP16 / BF16 (measured)

GPU — Throughput (samples/sec)

Batch	FP32	FP16	BF16	FP16 Speedup	BF16 Speedup
32	2,908	7,141	7,228	2.46×	2.49×

Batch	FP32	FP16	BF16	FP16 Speedup	BF16 Speedup
64	2,725	6,954	6,905	2.55×	2.53×
128	2,649	5,974	5,742	2.26×	2.17×
256	2,600	5,903	5,646	2.27×	2.17×
512	2,572	5,831	5,601	2.27×	2.18×
1,024	2,537	5,499	5,250	2.17×	2.07×

GPU — Peak VRAM (MB) — all batch sizes

Batch	FP32 (MB)	FP16 (MB)	BF16 (MB)	FP16 saving	BF16 saving
32	296	233	232	21.2%	21.6%
64	531	400	400	24.6%	24.6%
128	980	715	716	27.0%	26.9%
256	1,852	1,319	1,319	28.8%	28.8%
512	3,597	2,549	2,549	29.1%	29.1%
1,024	7,087	5,033	5,033	29.0%	29.0%

VRAM reduction is identical for FP16 and BF16 at every batch size (their lines overlap in Chart 3). The saving is batch-dependent: **~21% at batch=32**, growing to **~29% at batch≥256** where it stabilises. At small batch sizes, the fixed FP32 overhead (optimizer states, which are not halved under autocast) dilutes the activation-level savings; as batch size grows, activation memory dominates and the saving approaches ~29%.

TPU — Throughput (samples/sec)

FP16 is not natively supported on TPU v5litepod; only FP32 and BF16 are tested.

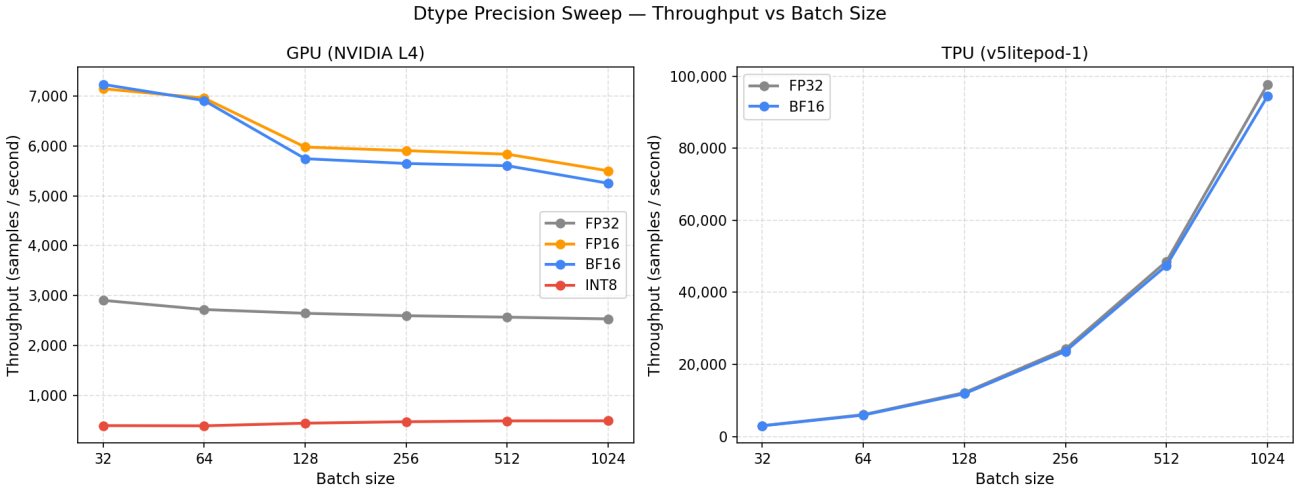
Batch	FP32	BF16	BF16 Speedup
32	3,006	2,965	0.986×
64	6,070	5,938	0.978×
128	12,175	11,873	0.975×
256	24,297	23,650	0.973×

Batch	FP32	BF16	BF16 Speedup
512	48,614	47,458	0.976×
1,024	97,749	94,522	0.967×

BF16 is **1.4–3.3% slower** than FP32 on the TPU (average ~2.4% across batch sizes), indicating a systematic overhead rather than measurement noise.

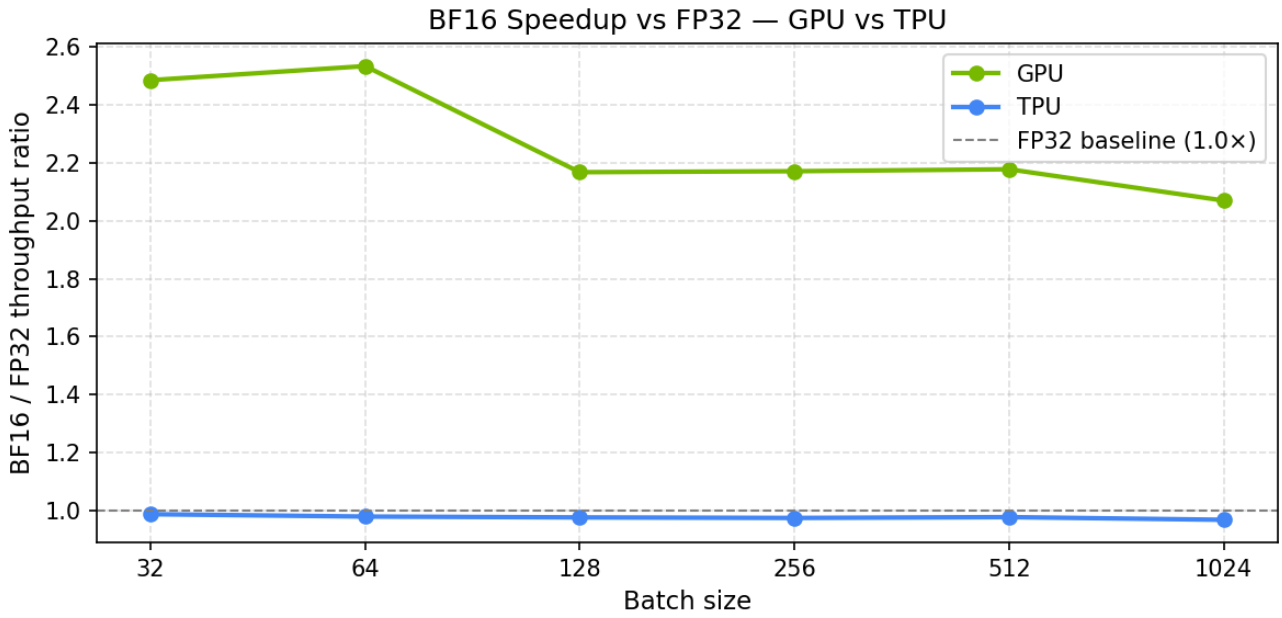
Charts (FP32 / FP16 / BF16)

Chart 1 — Throughput vs batch size (all dtypes)



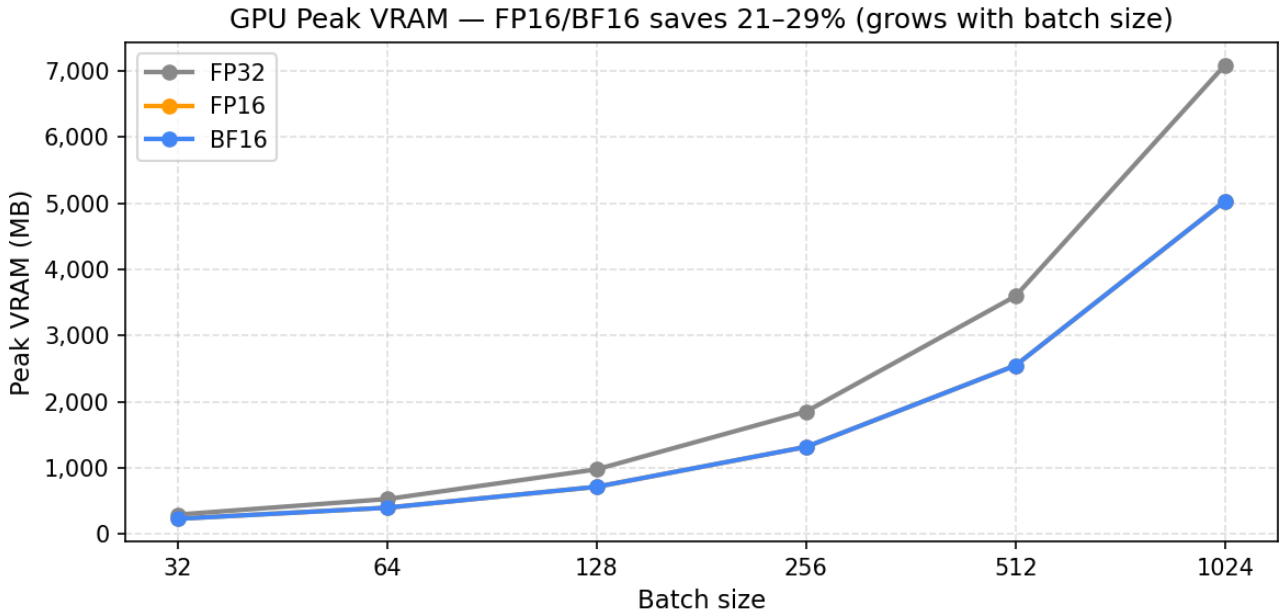
GPU lines separate into two bands — a lower FP32 band (~2,500–2,900 samples/sec, roughly flat) and an upper FP16/BF16 band (~5,250–7,200 samples/sec) that declines gently from batch=32 to batch=1024. The upper band peaks at batch=32–64 (~7,100–7,200 s/s) and converges toward ~5,300–5,500 s/s at batch=1024. TPU lines rise steeply and linearly with batch; at batch=1024 the TPU FP32 line reaches ~97,700 samples/sec, dwarfing all GPU lines. TPU FP32 and TPU BF16 are nearly coincident, making the 1.4–3.3% BF16 regression visible only at larger batches.

Chart 2 — BF16 / FP32 speedup ratio (GPU vs TPU)



The GPU BF16 speedup line ranges from **~2.5×** at batch=32–64 down to **~2.1×** at batch=1024, with a pronounced dip at batch=128 before stabilising. The TPU BF16 line sits just below **1.0×** and also stays flat — BF16 is consistently slower regardless of batch size. The chart makes the contrast between devices immediate.

Chart 3 — GPU peak VRAM by dtype



Three lines track peak VRAM from batch=32 to batch=1024. FP32 is the highest. FP16 and BF16 are coincident throughout. The relative gap between FP32 and the reduced-

precision lines narrows from ~21% at batch=32 to ~29% at batch≥256 as activation memory becomes the dominant term and fixed FP32 overhead is diluted.

Analysis: FP32 / FP16 / BF16

Why the GPU gains 2×+ from lower precision

NVIDIA Ada Lovelace GPUs contain Tensor Cores that execute FP16 and BF16 matrix multiplications at roughly twice the FLOP/s of FP32 operations. A Transformer block's forward and backward passes are dominated by matrix multiplications, so the Tensor Core advantage maps almost directly onto end-to-end training throughput.

The VRAM reduction (21–29%, batch-dependent) primarily reflects halved activation memory in the forward pass under autocast. Optimizer states (Adam momentum and variance) are kept in FP32 and are not reduced; this is why the saving is lower at small batch sizes (where fixed FP32 state dominates) and converges to ~29% at large batch (where activations dominate). In practice this enables a larger effective batch size or deeper model, especially at batch≥64.

Why FP16 requires a GradScaler but BF16 does not

FP16's narrow dynamic range (min nonzero $\sim 6\times10^{-5}$) causes gradients to flush to zero. `GradScaler` multiplies the loss by a large constant before backward and scales back after, monitoring for overflow. BF16 retains FP32's full 8-bit exponent range — no scaling needed. For new work on modern hardware (Ampere+), BF16 is simpler and numerically safer.

Why the TPU does not benefit from BF16

The v5litepod BF16 throughput is **1.4–3.3% lower** (average ~2.4%) than FP32 despite dedicated BF16 MXUs. Likely causes:

1. **Compile-time optimisation path.** XLA may choose a more aggressively fused FP32 kernel that happens to outperform the BF16 path for this model shape (d_model=512, seq_len=128). The MXU advantage is most pronounced at large matrix dimensions.
2. **Mixed-precision overhead.** Autocast inserts cast operations between FP32 compute (layer norm, loss) and BF16 compute (matrix multiplications). These casts add latency in the XLA graph that may exceed the savings for this model size.

The practical conclusion: FP32 is the correct default on TPU v5litepod for this family of model sizes. Benchmark BF16 per-model rather than assuming it helps.

INT8 — Experimental results

GPU INT8 (dynamic quantisation — CPU only)

`torch.ao.quantization.quantize_dynamic` in PyTorch applies INT8 to linear layer weights but executes on CPU rather than GPU. This is a limitation of the current `quantize_dynamic` API — it does not route through CUDA INT8 kernels.

Measured (CPU inference, not GPU INT8):

Batch	CPU INT8 (samples/s)	GPU FP32 (samples/s)	Ratio
32	402	2,908	0.138×
64	398	2,725	0.146×
128	451	2,649	0.170×
256	480	2,600	0.185×
512	497	2,572	0.193×
1,024	498	2,537	0.196×

CPU dynamic-quant throughput is **5–7× slower** than GPU FP32 — these are CPU numbers, not GPU INT8. To exercise the L4's 242 INT8 TOPS, use `torch.ao.quantization.prepare` with a CUDA-aware backend or `torch.quantization.quantize_fx` with device=cuda.

Expected GPU INT8 throughput (not yet measured): With proper CUDA INT8 kernels the L4 should deliver ~1.5–2× over FP32 for this model size (~3,800–5,100 samples/sec), consistent with the 2× spec ratio between INT8 TOPS and FP32 FLOPS.

TPU INT8

Casting weights to `torch.int8` and running inference on TPU v5litepod errored in torch_xla 2.9.0 with: "data set to a tensor that requires gradients must be floating point or complex dtype." The INT8 MXU path (786 TOPS = 2× BF16 spec) was not exercised.

Key Takeaways

- **On the GPU (NVIDIA L4), FP16 and BF16 each provide a ~2.1–2.5× throughput gain over FP32.** The speedup peaks at batch=32–64 (~2.5×) and settles to ~2.1–2.3× at batch≥128. Both formats simultaneously reduce peak VRAM by ~21–29% (batch-dependent), enabling larger batches or bigger models at no cost.

- **BF16 is simpler than FP16 in practice.** BF16's wider exponent range matches FP32's dynamic range, so gradient underflow does not occur and no `GradScaler` is needed.
- **On the TPU v5litepod, BF16 is 1.4–3.3% slower than FP32** (average ~2.4%) across all tested batch sizes. The BF16 MXU advantage does not materialise for this model shape. **Use FP32 on this TPU configuration until benchmarked otherwise.**
- **The TPU's raw throughput advantage at large batches is substantial regardless of dtype.** At batch=1024, FP32 on the TPU reaches 97,749 samples/sec vs 2,537 on the GPU — a ~39× gap. The dtype choice is a secondary concern on the TPU; batch size is the dominant lever.
- **GPU INT8 (`quantize_dynamic`) runs on CPU in PyTorch, not on the GPU's INT8 Tensor Cores.** The measured ~400–498 samples/sec is a CPU baseline, not GPU INT8 capability. A CUDA-native INT8 path would require a different quantisation API.
- **TPU INT8 is not accessible via torch_xla 2.9.0** with the weight-casting approach. The v5e INT8 MXU (786 TOPS) remains unmeasured in this session.

Decision Rule from This Session

- **GPU workloads:** always use BF16 (or FP16 with a GradScaler on older hardware). The ~2× throughput and 21–29% VRAM savings are effectively free.
- **TPU workloads:** benchmark before assuming BF16 helps. On v5litepod (d_model=512, seq_len=128), FP32 is faster. Revisit if using v4 TPUs, larger model dimensions, or dedicated BF16-only inference paths.
- **INT8 for inference:** use when throughput matters more than numeric precision. For GPU INT8, use a CUDA-aware quantisation path (not `quantize_dynamic`). For TPU INT8, a newer torch_xla version or JAX is needed to exercise the MXU INT8 path.

Session 6: Framework Interoperability — JAX/Flax and PyTorch/XLA on Shared Hardware

Overview

Sessions 1–5 reached the GPU and TPU through PyTorch and PyTorch/XLA. Session 6 asks whether the ML framework matters, or whether the XLA compiler decides everything underneath.

Two production ML frameworks — **JAX/Flax** and **PyTorch/XLA** — are compared on the same hardware, benchmarking the same Transformer encoder architecture. Both ultimately compile to XLA HLO before executing. The question is whether the two front-ends generate equivalent programs, and what the performance consequences are if they don't.

This session illustrates a broader point about the ML hardware stack: **shared hardware does not mean equivalent execution**. The same accelerator can behave very differently depending on how a framework traces, compiles, and dispatches operations. Understanding this is critical for practitioners choosing or migrating between frameworks.

The same Transformer encoder architecture used in Sessions 1–5 is re-implemented in Flax/Linen (`flax_model.py`) and benchmarked across the same batch sweep as Session 1 ([4, 8, 16, 32, 64, 128, 256, 512, 1024]).

Notebooks: `session_6/`

Hardware

Same devices as Sessions 1–5. See `session_1.md` for full hardware specs.

Device	Memory	Note
NVIDIA L4 (GPU)	23.7 GB GDDR6	JAX uses CUDA XLA backend (not cuDNN-native like PyTorch)
TPU v5litepod-1	16 GB HBM2	JAX and PyTorch/XLA both compile to XLA HLO here

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Software Environment

Component	GPU	TPU
Python	3.12.12	3.12.12
JAX	0.9.0.1	0.9.0.1
Flax	0.12.4	0.12.4
Backend	`jax[cuda12]`	`jax[tpu]`
Run timestamp	2026-02-28T00:38 UTC	2026-02-28T00:36 UTC

Benchmark Configuration

Parameter	Value
Model	`FlaxTransformerBlock` (matches `BenchmarkTransformerBlock` architecture)
`D_MODEL`	512
`N_HEAD`	8
`DIM_FEEDFORWARD`	2048
`SEQ_LEN`	128 (fixed)
`BATCH_SIZES`	4, 8, 16, 32, 64, 128, 256, 512, 1024
Steps (batch ≤ 256)	50 (+ 5 warmup)
Steps (batch ≥ 512)	20 (+ 5 warmup) — GPU only, adaptive to prevent hang
Loop	forward → backward → Adam step (`jax.value_and_grad` + `optax.adam`)
Metric	throughput (samples/sec), latency (ms/step), first-call compilation time (s)

Key difference from PyTorch sessions

JAX uses **functional parameter management**: model weights are a pytree returned by `model.init()` and passed explicitly into `model.apply()` on every call. There is no stateful model object. The `jax.jit`-compiled `train_step` function captures the model and optimizer as Python closures; JAX traces through both the forward pass and the `jax.value_and_grad` backward pass in a single XLA program.

Architecture equivalence

`FlaxTransformerBlock` in `flax_model.py` replicates `BenchmarkTransformerBlock` exactly:

PyTorch (`transformer_block.py`)

```
x = norm1(x + attn(x, x, x)[0])
x = norm2(x + ff(x))
```

Flax (`flax_model.py`)

```
attn_out = MultiHeadDotProductAttention(...)(x, x)
x = LayerNorm()(x + attn_out)
ff = Dense(DIM_FEEDFORWARD)(x) → gelu → Dense(D_MODEL)(x)
x = LayerNorm()(x + ff)
```

Both are **Post-LayerNorm** (norm applied after residual). The feedforward uses GELU. The attention block is self-attention (query = key = value = x) with `qkv_features = out_features = D_MODEL`.

Results

GPU — Throughput (samples/sec) and Compilation Time

JAX/Flax results vs PyTorch Session 1 baseline:

Batch	JAX/Flax	PyTorch (S1)	JAX/PT ratio	Compile time
4	3,186	1,388	2.30×	15.6 s
8	4,409	2,521	1.75×	18.0 s
16	5,946	2,866	2.07×	19.5 s

Batch	JAX/Flax	PyTorch (S1)	JAX/PT ratio	Compile time
32	6,070	2,895	2.10×	22.6 s
64	5,304	2,760	1.92×	18.8 s
128	4,823	2,688	1.79×	19.5 s
256	4,660	2,653	1.76×	20.4 s
512	4,571	2,618	1.75×	22.0 s
1,024	4,463	2,575	1.73×	26.5 s

Unexpected finding: JAX/Flax is **1.7–2.3× faster** than PyTorch native on GPU across all batch sizes. This reverses the pre-run expectation. See Analysis below.

TPU — Throughput (samples/sec) and Compilation Time

JAX/Flax results vs PyTorch/XLA Session 1 baseline:

Batch	JAX/Flax	PyTorch/XLA (S1)	JAX/PT ratio	Compile time
4	8,454	370	22.8×	2.5 s
8	15,867	745	21.3×	3.6 s
16	26,983	1,497	18.0×	3.8 s
32	38,689	2,984	13.0×	3.9 s
64	39,160	5,968	6.6×	2.8 s
128	36,405	11,900	3.1×	3.3 s
256	30,820	23,862	1.3×	3.3 s
512	28,620	47,580	**0.60×**	3.5 s
1,024	28,418	94,795	**0.30×**	3.5 s

Two unexpected findings on TPU: 1. JAX is **dramatically faster at small batch sizes** — up to 22.8× at batch=4 — but **PyTorch/XLA wins decisively at large batch sizes** (batch ≥ 512). 2. JAX TPU throughput **plateaus around batch=64** (~39k samples/sec) and then **declines**, while PyTorch/XLA scales linearly all the way to batch=1024 (95k samples/sec).

The crossover is between batch=256 (JAX still ahead 1.3×) and batch=512 (PyTorch/XLA 1.67× faster).

Analysis

Why JAX outperforms PyTorch on GPU (whole-graph fusion)

JAX's `jax.jit` compiles the entire train step — forward pass, `jax.value_and_grad` backward, and the `optax.adam` optimizer update — into a single XLA program. XLA can then fuse adjacent operations across this full graph: it eliminates intermediate tensor materializations, reorders memory-bound ops, and generates a single kernel sequence that minimises GDDR6 round-trips.

PyTorch's eager mode dispatches each operation as a separate kernel call: the forward pass, then the backward pass (autograd), then each optimizer parameter update. Each call crosses the Python–CUDA boundary and potentially writes an intermediate result to GDDR6 before the next call reads it back. For a single Transformer block, this per-operation overhead is the dominant cost at small batch sizes — not raw compute.

The XLA advantage declines from 2.30× at batch=4 to 1.73× at batch=1024 as the GPU becomes more compute-saturated and the overhead is diluted.

Why JAX plateaus and declines on TPU at large batch sizes

On TPU, both frameworks compile to XLA HLO — but they generate different programs.

PyTorch/XLA traces through the PyTorch autograd graph and emits a series of matrix operations. Its throughput scales perfectly linearly with batch size (Session 1 data), indicating the MXU is purely compute-bound and XLA is efficiently pipelining operations through the systolic array.

JAX/Flax compiles the full train step (including Adam optimizer) in a single `jax.jit` trace. The resulting XLA program materialises all intermediate gradient and optimizer state tensors in a single step. At large batch sizes, the gradient tensors (same shape as the activations, scaling with batch size) and the Adam moment updates create HBM pressure that the JAX-generated program cannot fully hide behind compute.

The plateau at batch=64 (~39k samples/sec) and subsequent decline suggests JAX's XLA program becomes **HBM bandwidth-bound at large batch sizes** on the v5litepod, despite the MXU still being available. PyTorch/XLA's approach — which separates the forward, backward, and optimizer steps into a sequence of `mark_step()` flushes — allows XLA to compile smaller, more focused kernels that pipeline better at scale.

What this reveals about framework-hardware abstraction

Both frameworks claim XLA as their backend. Both run on the same physical hardware. Yet they deliver up to 22.8× different throughput (batch=4 on TPU) and reverse their relative ranking depending on batch size. The key lesson is:

"Same backend" does not mean "same program." The XLA HLO emitted by JAX and PyTorch/XLA differs in: - Graph granularity (JAX: one program per train step; PyTorch/XLA: one program per flush) - Memory layout decisions made during lowering - Which operations XLA can fuse once it receives the graph

Practitioners choosing between frameworks for TPU workloads should benchmark both at their actual batch size — the decision depends on where on the batch-size axis they operate.

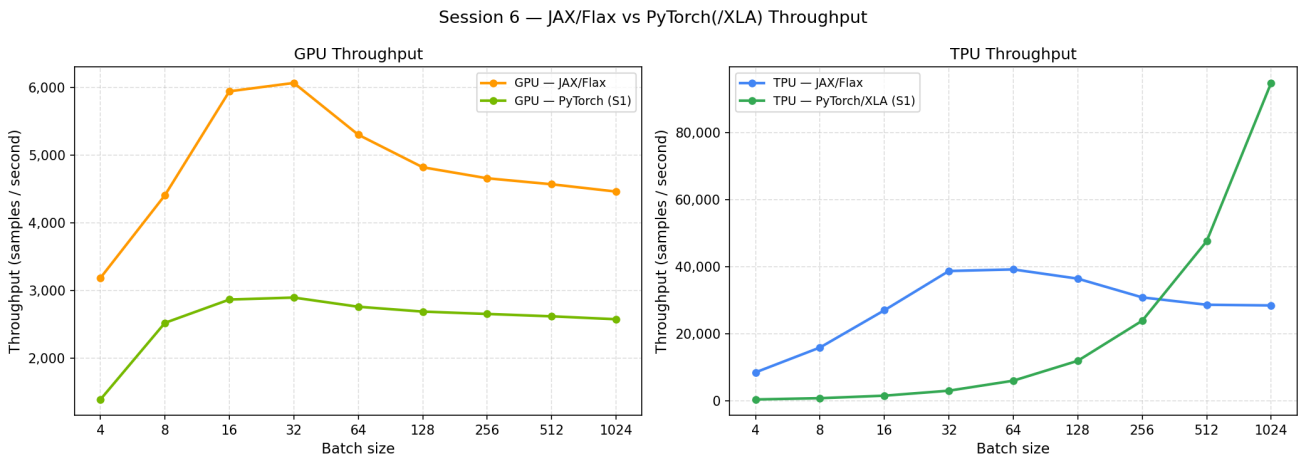
Compilation cost comparison

Device	JAX compile (per shape)	PyTorch equivalent
GPU	15–27 s	~0 ms (cuDNN kernel lookup)
TPU	2.5–3.9 s	First `mark_step()`: ~seconds

JAX TPU compilation is ~7× faster than JAX GPU compilation, reflecting that XLA is more native to TPU and the compilation pipeline is more optimised for that target.

Charts

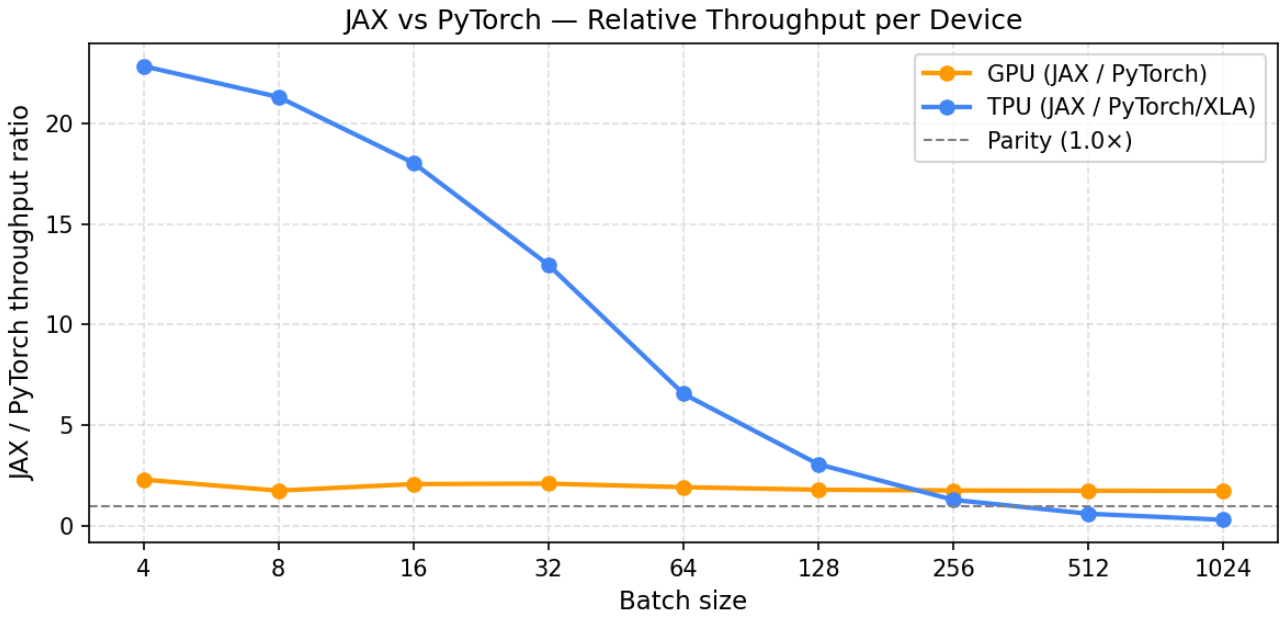
Chart 1 — Throughput vs batch size (JAX/Flax vs PyTorch)



GPU panel: JAX/Flax sits above the PyTorch line across all batch sizes. JAX peaks at batch=32 (~6,070 s/s) then declines to ~4,460 s/s at batch=1024 — a characteristic of XLA whole-graph fusion hitting bandwidth limits at larger footprints. PyTorch is roughly flat from batch=16 onward (~2,600–2,900 s/s).

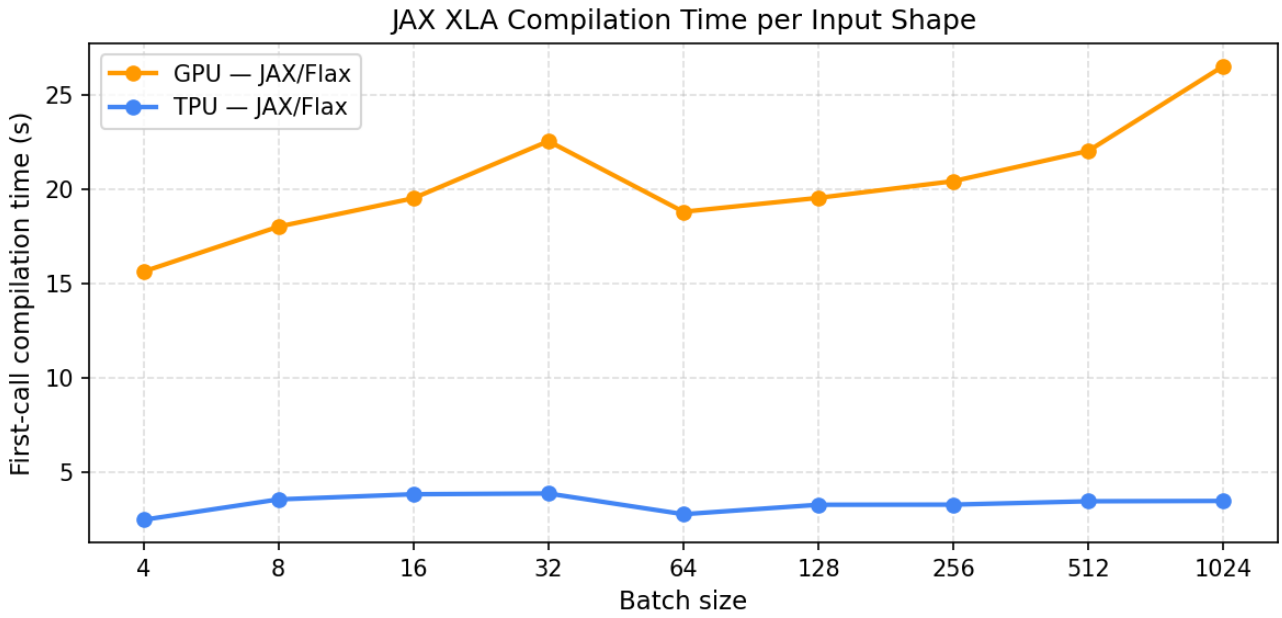
TPU panel: JAX peaks at batch=32–64 (~39k samples/sec) and declines. PyTorch/XLA rises linearly to ~95k samples/sec at batch=1024. The two lines cross between batch=256 and batch=512.

Chart 2 — JAX / PyTorch throughput ratio per device



GPU: Ratio stays above 1.0× across all batch sizes (JAX always faster on GPU). **TPU:** Ratio starts at 22.8× (batch=4), falls through 1.0× at batch=256–512, reaches 0.30× at batch=1024 (PyTorch/XLA 3.3× faster).

Chart 3 — Compilation time per input shape



GPU compilation: 15–27 s (grows slightly with batch size — larger input shapes take longer to lower to PTX). TPU compilation: 2.5–3.9 s (relatively flat — the XLA→TPU lowering is more efficient than XLA→CUDA).

Key Takeaways

- **On GPU, JAX/Flax is 1.7–2.3× faster than PyTorch native.** XLA's whole-graph compilation of the full train step eliminates intermediate tensor materialisation costs that PyTorch eager dispatch incurs. The advantage is larger at small batch sizes where bandwidth overhead dominates.
- **On TPU at small batch sizes, JAX is dramatically faster than PyTorch/XLA** — up to 22.8× at batch=4. The JAX XLA program is far more aggressive about fusing small-batch operations than PyTorch/XLA's `mark_step()` approach.
- **On TPU at large batch sizes, PyTorch/XLA wins decisively.** JAX throughput plateaus at ~39k samples/sec (batch=64) and declines to 28k at batch=1024, while PyTorch/XLA scales linearly to ~95k. The crossover is around batch=300–400.
- **"Both use XLA" does not mean "equivalent performance."** The frameworks generate meaningfully different XLA programs. The performance crossover is batch-size-dependent, and the winning framework can differ by up to 22×.
- **JAX TPU compilation is 2.5–3.9 s per shape** — roughly 7× faster than JAX GPU (15–27 s). PyTorch/XLA first-step compilation is in a similar range on TPU.

Decision Rule from This Session

Scenario	Recommendation	Reason
GPU, existing PyTorch code	**Stay on PyTorch**	JAX faster, but migration cost exceeds benefit for existing codebases
GPU, new project, fixed batch	**Consider JAX**	1.7–2.3× throughput gain is real; compilation cost amortises quickly
TPU, large batch (≥ 512)	**PyTorch/XLA**	Scales linearly to 95k samples/sec; JAX declines to 28k
TPU, small batch (≤ 64)	**JAX**	Up to 22.8× faster; PyTorch/XLA is slow here
TPU, batch ≈ 256	**Either**	Performance is within 1.3× — ergonomics decides